Theory and Phenomenology of Superconductivity Homework 5

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Problem 1. • Consider a system consisting of N spin-$\frac{1}{2}$ particles, each of which can be in one of two quantum states, namely $\uparrow$ and $\downarrow$. In presence of a magnetic field B, the energy of a spin in a $\uparrow$ / $\downarrow$ state is $\epsilon = \pm\mu_B B/2$ where μ_B is the magnetic moment. Show that the partition function is

$$Z = 2^N \cosh^N \left(\frac{\mu_B B}{2k_B T} \right),$$

with $1/\beta = k_B T$ in the canonical ensemble. Find the average energy E and entropy S. Compute both quantities at zero temperature and $T \to \infty$.

Proof. The partition function of a single particle is

$$\begin{aligned} Z_1 &= \sum_{s \in \{\uparrow,\downarrow\}} e^{-\beta H(s)} \\ &= e^{-\beta\epsilon_\uparrow} + e^{-\beta\epsilon_\downarrow} \\ &= e^{\frac{\beta\mu_B B}{2}} + e^{-\frac{\beta\mu_B B}{2}} \\ &= 2\cosh \frac{\mu_B B}{2k_B T}. \end{aligned}$$

The partition function of N particles is the product of their partition functions; in this case, it is simply

$$Z = Z_1^n = 2^N \cosh^N \left(\frac{\mu_B B}{2k_B T} \right).$$

The average energy is given by

$$\begin{aligned} \langle E \rangle &= -\frac{\partial \ln Z}{\partial \beta} \\ &= -\frac{\partial}{\partial \beta} N \ln \left[2\cosh \left(\frac{\beta\mu_B B}{2} \right) \right] \\ &= -N \frac{\partial}{\partial \beta} \ln\cosh \frac{\beta\mu_B B}{2} \end{aligned}$$

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$$
\begin{aligned}
&= -\frac{N}{\cosh \frac{\beta \mu_B B}{2}} \left(\sinh \frac{\beta \mu_B B}{2} \right) \left(\frac{\mu_B B}{2} \right) \\
&= -\frac{N \mu_B B}{2} \tanh \frac{\beta \mu_B B}{2} \\
&= -\frac{N \mu_B B}{2} \tanh \left(\frac{\mu_B B}{2 k_B T} \right)
\end{aligned}
$$

The Helmholtz free energy is given by

$$
F(T, V, N) = -k_B T \ln Z.
$$

The entropy can be computed from the Helmholtz free energy as

$$
S = -\frac{\partial F}{\partial T}.
$$

This leads to the expression

$$
\begin{aligned}
\frac{\partial \ln Z}{\partial T} &= N \frac{\partial}{\partial T} \left(\ln \left[2 \cosh \left(\frac{\mu_B B}{2 k_B T} \right) \right] \right) \\
&= N \frac{\partial}{\partial T} \left(\ln \cosh \left(\frac{\mu_B B}{2 k_B T} \right) \right) \\
&= N \frac{1}{\cosh \frac{\mu_B B}{2 k_B T}} \left(\sinh \frac{\mu_B B}{2 k_B T} \right) \left(-\frac{\mu_B B}{2 k_B T^2} \right) \\
&= -N \frac{\mu_B B}{2 k_B T^2} \tanh \frac{\mu_B B}{2 k_B T}
\end{aligned}
$$

Then, we expand

$$
\begin{aligned}
S &= -\frac{\partial F}{\partial T} \\
&= k_B \ln Z + k_B T \frac{\partial}{\partial T} \ln Z \\
&= k_B N \ln \left[2 \cosh \left(\frac{\mu_B B}{2 k_B T} \right) \right] - \frac{N \mu_B B}{2T} \tanh \frac{\mu_B B}{2 k_B T}
\end{aligned}
$$

We note that

$$
\lim_{x \to \infty} \tanh(x) = 1, \qquad \lim_{x \to -\infty} \tanh(x) = -1, \qquad \tanh(0) = 0.
$$

Thus, the average energy $\langle E \rangle$ is, in the limits,

$$
\begin{aligned}
\langle E \rangle &\xrightarrow{T \to 0} -\frac{N \mu_B B}{2}, \\
\langle E \rangle &\xrightarrow{T \to \infty} 0.
\end{aligned}
$$

In the limit of $T \to \infty$, the second term in the entropy vanishes due to the $\frac{1}{T}$factor. Since $\cosh(0) = 1$, we are left with

$$S \xrightarrow{T\to\infty} = k_B N \ln 2.$$

The limit $T \to 0$ is the most subtle. We massage the entropy into a desired form by

$$\begin{aligned}
S &= k_B N \left\{ \ln \left[2\cosh\left(\frac{\mu_B B}{2k_B T}\right)\right] - \frac{\mu_B B}{2k_B T} \tanh \frac{\mu_B B}{2k_B T} \right\} \\
&= k_B N \left\{ \ln \left[2\cosh\left(\frac{\mu_B B}{2k_B T}\right)\right] - \ln \exp \left[\frac{\mu_B B}{2k_B T} \tanh \frac{\mu_B B}{2k_B T} \right] \right\} \\
&= k_B N \ln \frac{e^{\frac{\mu_B B}{2k_B T}} + e^{-\frac{\mu_B B}{2k_B T}}}{\left(e^{\frac{\mu_B B}{2k_B T}}\right)^{\tanh \frac{\mu_B B}{2k_B T}}} \\
&= k_B N \ln \frac{1 + e^{-\frac{\mu_B B}{k_B T}}}{\left(e^{\frac{\mu_B B}{2k_B T}}\right)^{\tanh\left(\frac{\mu_B B}{2k_B T}\right) - 1}}
\end{aligned}$$

Now all we need to do is to simplify

$$\begin{aligned}
\exp\left(\frac{\mu_B B}{2k_B T}\left(\tanh\left(\frac{\mu_B B}{2k_B T}\right) - 1\right)\right) &= \exp\left(\frac{\mu_B B}{2k_B T}\left(\frac{e^{\frac{\mu_B B}{k_B T}} - 1}{e^{\frac{\mu_B B}{k_B T}} + 1} - 1\right)\right) \\
&= \exp\left(\frac{\mu_B B}{2k_B T} \frac{-2}{e^{\frac{\mu_B B}{k_B T}} + 1}\right) \\
&\xrightarrow{T\to 0} 1
\end{aligned}$$

and hence

$$S \xrightarrow{T\to 0} k_B N \ln 1 = 0. \qquad \square$$

Problem 2.

- Compute the partition function of a quantum harmonic oscillator at frequency ω in the canonical ensemble. *Hint:* The energy levels are given by

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right),$$

with $n \in \mathbb{Z}$.

- A simple model of a solid can be made considering N atoms that vibrate all of them at the same frequency ω. Consider these vibrations as a harmonic oscillator. Show that at high temperatures, $k_B T \gg \hbar\omega$, one has a heat capacity

$$C_V = N k_B.$$

- Derive the limit also for low temperatures.

Proof. The partition function is given by

$$\begin{aligned}
Z &= \sum_n e^{-\beta E_n} \\
&= \sum_{n=0}^{\infty} e^{-\beta\hbar\omega\left(n+\frac{1}{2}\right)} \\
&= e^{-\frac{\beta\hbar\omega}{2}} \sum_{n=0}^{\infty} e^{-\beta\hbar\omega n} \\
&= e^{-\frac{\beta\hbar\omega}{2}} \frac{e^{\beta\hbar\omega}}{e^{\beta\hbar\omega}-1} \\
&= \frac{1}{e^{\frac{\beta\hbar\omega}{2}} + e^{-\frac{\beta\hbar\omega}{2}}} \\
&= \frac{1}{2\sinh\left(\frac{\beta\hbar\omega}{2}\right)}
\end{aligned}$$

We then determine the average energy of a single particle as in the previous problem

$$\begin{aligned}
U := \langle E \rangle &= -\frac{\partial \ln Z}{\partial \beta} \\
&= \frac{\partial}{\partial \beta} \ln\left[2\sinh\left(\frac{\beta\hbar\omega}{2}\right)\right] \\
&= \frac{\partial}{\partial \beta} \ln\left[\sinh\left(\frac{\beta\hbar\omega}{2}\right)\right] \\
&= \frac{1}{\sinh\left(\frac{\beta\hbar\omega}{2}\right)} \left[\cosh\left(\frac{\beta\hbar\omega}{2}\right)\right] \frac{\hbar\omega}{2} \\
&= \frac{\hbar\omega}{2} \coth\left(\frac{\beta\hbar\omega}{2}\right)
\end{aligned}$$

At high temperatures, β is low and we can make use of the well known expansion in small β

$$U = \frac{\hbar\omega}{2}\frac{2}{\beta\hbar\omega} + O(\beta) = k_B T + O\left(\frac{1}{T}\right).$$

By inspection, we see that the heat capacity $\frac{\partial U}{\partial T} = k_B$, and for N noninteracting harmonic oscillators, we simply multiply by N to get

$$C_V = N k_B.$$

In the low temperature limit, β is very high. We approximate

$$\cosh(x) \approx \frac{e^x}{2}$$

$$\sinh(x) \approx \frac{e^x}{2}$$

to get

$$U \approx \frac{\hbar\omega}{2}.$$

Since U does not depend on T in this limit, the heat capacity $\frac{\partial U}{\partial T}$ vanishes - without putting in any heat, it is possible to change the temperature. □

Problem 3.

- Consider the Gibbs entropy for a probability distribution $p(n)$,

$$S = -k_B \sum_n p(n) \ln p(n).$$

- Through the use of a Lagrange multiplier, show that when restricted to states of fixed energy E, the entropy is maximized by the microcanonical ensemble, in which all such states are equally likely. Further show that in this case, the Gibbs entropy coincides with the Boltzmann entropy. *Hint:* Recall that probabilities are positive and constrained to sum up to 1.

- Show that at fixed average energy, i.e.: $\langle E \rangle = \sum_n p(n) E_n$, the entropy is maximized by the canonical ensemble. Moreover, show that the Lagrange multiplier imposing the constraint is proportional to the inverse of temperature, β. Check that maximizing the entropy is equivalent to minimizing the free energy.

Proof. (a) The constraint equation is given by

$$\sum_n p(n) = 1.$$

Hence, the (constrained) maximization of the entropy can be replaced by an unconstrained maximization over the functional

$$S' = -k_B \sum_n p(n) \ln p(n) + \lambda \left(\sum_n p(n) - 1 \right).$$

We extremize this by taking the partial derivatives. The partial derivative with respect to an occupation $p(k)$ is given by

$$\frac{\partial S'}{\partial p(k)} = -k_B \left(\ln p(k) + 1 \right) + \lambda = 0$$

and the partial derivative with respect to λ returns the constraint equation

$$\sum_n p(n) = 1$$

as expected. Rearranging the partial derivative with respect to $p(k)$ yields

$$\ln p(k) = \frac{\lambda}{k_B} - 1.$$

Since this must be true for all k, it tells us that $\ln p(k)$ is the same for all k. Since ln is monotonic, then $p(k)$ must be the same for all k. In this case, the normalisation condition yields, for a total of N microstates,

$$p(k) = \frac{1}{N} \ \forall k.$$

The Gibbs entropy is given by

$$\begin{aligned} S &= -k_B \sum_{n=1}^{N} \frac{1}{N} \ln \frac{1}{N} \\ &= k_B \ln N \end{aligned}$$

This is, by definition, the Boltzmann entropy.

(b) We follow the same procedure, defining instead the functional

$$S' = -k_B \sum_n p(n) \ln p(n) + \lambda \left(\sum_n p(n) E_n - U \right) + \eta \left(\sum_n p(n) - 1 \right)$$

where we denote the average energy by $\langle E \rangle =: U$. Then, the derivatives yield

$$\begin{aligned} \frac{\partial S'}{\partial p(k)} &= -k_B(\ln p(k) + 1) + \lambda E_k + \eta = 0 \\ \frac{\partial S'}{\partial \lambda} &= \sum_n p(n) E_n - U = 0 \\ \frac{\partial S'}{\partial \eta} &= \sum_n p(n) - 1 = 0 \end{aligned}$$

We proceed in a manner analogous to the previous part: By solving the equation for $p(k)$. We have

$$\ln p(k) + 1 = \frac{1}{k_B} (\eta + \lambda E_k)$$

and

$$p(k) = e^{-1} e^{\frac{1}{k_B}(\eta + \lambda E_k)}$$

$$= e^{-1} e^{\eta/k_B} e^{\lambda E_k}$$

From this, we see by inspection that λ is proportional to the inverse temperature β.

(c) The Helmholtz free energy is given by

$$F = U - TS.$$

Since U and T are fixed in the canonical ensemble, minimizing S is equivalent to maximizing F. □

Problem 4. Bose-Einstein condensation was obtained experimentally for the first time in 1995 by confining particles in a harmonic potential. Suppose you have particles in a potential $V(\mathbf{r}) = \frac{1}{2} m\omega^2 r^2$. Then, the particle states are characterized by three quantum numbers, n_x, n_y, n_z, all integers and greater or equal to zero. The energy of each state is given by

$$\epsilon_{n_x,n_y,n_z} = \hbar\omega(n_x + n_y + n_z).$$

Notice that the zero energy level has been redefined so that the lowest energy state (the ground state) has energy 0.

- Show that the grand canonical partition function can be written as

$$\ln Z_G = -\sum_{n_x,n_y,n_z} \ln\left(1 - z e^{-\beta\hbar\omega(n_x+n_y+n_z)}\right),$$

  where $z = e^{\mu\beta}$ is the fugacity.

- Introduce $n = n_x + n_y + n_z$ and rewrite the sum over states as the sum over n with multiplicity $\Omega(n)$. Compute the number of states of a particle $\Omega(n)$ at a given energy $E_n = \hbar\omega n$ (notice that it is equivalent as, for instance, to distribute n equal books in 3 indistinguishable boxes).

- Which condition must be satisfied in order to approximate the sum by an integral? Approximate the sum in the grand canonical partition function by an integration

assuming that the dominant contributions are from terms with $n \gg 1$. $\ln Z_G$ should be written in terms of $\mathrm{Li}_s(z)$ defined as

$$\mathrm{Li}_s(z) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x/z - 1}\, dx,$$

with Γ is the Gamma function. $\mathrm{Li}_s(z)$ is the polylogarithm of order s. It will become useful that $\mathrm{Li}_s(1) = \zeta(s)$, where ζ is the Riemann Zeta function and $z\mathrm{Li}_s'(z) = \mathrm{Li}_{s-1}(z)$.

- Find the expected number of particles $N = \langle N \rangle$ expressed in terms of the fugacity. (Notice that the condensate is a particular limit when $z \to 1$.) Identify the occupation of the ground state and the occupation of excited states.

- Introduce $0 < f \leq 1$ to represent how many particles are in the lowest energy state, such that $z = 1 - 1/fN$. Discuss the thermodynamic limit and find the critical temperature T_c. The critical temperature is where the ground state starts to become occupied when lowering the temperature.

- Calculate the internal energy and specific heat of the gas of temperature below the critical temperature.

Proof. (a) The grand canonical partition function is given by

$$\begin{aligned} Z_G &= \sum e^{-\beta(H-\mu N)} \\ &= \sum_{n_x,n_y,n_z=0}^{\infty} \sum_{N(n_x,n_y,n_z)=0}^{\infty} e^{-\beta N(n_x,n_y,n_z)(\hbar\omega-\mu)(n_x+n_y+n_z)} \\ &= \sum_{n_x,n_y,n_z=0}^{\infty} \sum_{N(n_x,n_y,n_z)=0}^{\infty} \left(e^{-\beta(\hbar\omega-\mu)(n_x+n_y+n_z)}\right)^{N(n_x,n_y,n_z)} \\ &= \sum_{n_x,n_y,n_z=0}^{\infty} \frac{1}{1 - e^{-\beta(\hbar\omega-\mu)(n_x+n_y+n_z)}} \end{aligned}$$

where $N(n_x, n_y, n_z)$ is the number of particles in the state (n_x, n_y, n_z). Notably, n_x, n_y and n_z represent just energy states, rather than the number of bosons in a state. Taking the logarithm yields

$$\ln Z_G = -\sum_{n_x,n_y,n_z=0}^{\infty} \ln\left[1 - ze^{-\beta\hbar\omega(n_x+n_y+n_z)}\right].$$

(b) Clearly, the partition function is given by

$$\ln Z_G = \sum_{n=0}^{\infty} \sum_{n_x+n_y+n_z=n} \ln\left(1 - ze^{-\beta\hbar\omega(n_x+n_y+n_z)}\right).$$

Since the expression inside the sum only depends on n, the inner sum reduces to simply counting $\Omega(n)$. We begin by choosing any number between 0 and n for n_x. In general, after choosing n_x, we will then have $n - n_x + 1$ possibilities for n_y. This fixes n_z. Thus,

$$\Omega(n) = \sum_{n_x=0}^{n} (n - n_x + 1) = \frac{(n+1)(n+2)}{2}.$$

Thus the partition function is

$$\ln Z_G = \sum_{n=0}^{\infty} \Omega(n) \ln\left(1 - ze^{-\beta\hbar\omega n}\right) = \sum_{n=0}^{\infty} \frac{(n+1)(n+2)}{2} \ln\left(1 - ze^{-\beta\hbar\omega n}\right).$$

(c) In general, a sum can be approximated by an integral when it does not change much over an interval of length 1. This occurs when the energy spacing between the levels is small, i.e. $\beta\hbar\omega \leq 1$.

(d) For high n, the prefactor simplifies to $\frac{n^2}{2}$. Thus, we approximate the partition function using an integral

$$\begin{aligned}
\ln Z_G &\approx \int_0^{\infty} \frac{n^2}{2} \ln\left(1 - ze^{-\beta\hbar\omega n}\right) \mathrm{d}n \\
&= \left[\frac{n^3}{6} \ln\left(1 - ze^{-\beta\hbar\omega n}\right)\right]_0^{\infty} - \int_0^{\infty} \frac{n^3}{6} \frac{1}{1 - ze^{-\beta\hbar\omega n}} (-ze^{-\beta\hbar\omega n})(-\beta\hbar\omega)\, \mathrm{d}n
\end{aligned}$$

We must show that the constant part vanishes. Clearly the lower limit vanishes. The upper limit is a bit more subtle: We expand

$$\ln(1 - x) \approx -x$$

and note that the exponential prefactor now standing in front of the polynomial forces the term to vanish. Thus, we have

$$\begin{aligned}
\ln Z_G &= -\frac{\beta\hbar\omega}{6} \int_0^{\infty} \frac{n^3}{\frac{1}{z}e^{\beta\hbar\omega n} - 1}\, \mathrm{d}n \\
&= -\frac{\beta\hbar\omega}{6} \int_0^{\infty} \frac{1}{(\beta\hbar\omega)^4} \frac{(\beta\hbar\omega n)^3}{\frac{1}{z}e^{\beta\hbar\omega n} - 1}\, \mathrm{d}(\beta\hbar\omega n) \\
&= \frac{1}{6(\beta\hbar\omega)^3} (\Gamma(4)\mathrm{Li}_4(z)) \\
&= \frac{1}{(\beta\hbar\omega)^3} \mathrm{Li}_4(z)
\end{aligned}$$

(e) The grand canonical potential is given by

$$J = -\frac{1}{\beta} \ln Z_G.$$

We can then determine the expectation value of the particle number as

$$\langle N \rangle = -\left(\frac{\partial J}{\partial \mu}\right)_T.$$

This gives us

$$\begin{aligned}
\langle N \rangle &= \frac{1}{\beta} \frac{\partial}{\partial \mu} \ln Z_G \\
&= \frac{1}{\beta} \frac{1}{\beta \hbar \omega} \frac{\partial}{\partial \mu} \mathrm{Li}_4(z) \\
&= \frac{1}{\beta} \frac{1}{\beta \hbar \omega} \left(\frac{\partial}{\partial z} \mathrm{Li}_4(z)\right) \frac{\partial z}{\partial \mu}
\end{aligned}$$

The first derivative is clear:

$$\frac{\partial z}{\partial \mu} = z\beta.$$

For the second, we use the hint to get

$$\begin{aligned}
\langle N \rangle &= \frac{1}{\beta(\beta\hbar\omega)^3} \beta z \mathrm{Li}_4'(z) \\
&= \frac{1}{(\beta\hbar\omega)^3} \mathrm{Li}_3(z)
\end{aligned}$$

Anticipating a transition when lowering temperature, we analyse what happens when we lower the temperature while keeping the number of particles constant. As temperature decreases, the polylogarithmic function must increase to keep the number of particles constant. Hence, z increases and thus μ must increase. However, the maximum value of μ is 0 - at this value of μ, we have $z = 1$ and $\mathrm{Li}_3(1) = \zeta(3) \approx 1.2$.

Since we can decrease T arbitrarily, but cannot increase $\mathrm{Li}_3(z)$ arbitrarily to compensate, we must conclude that we have missed out something in our analysis. This is the assumption that there are no macroscopically occupied states. Thus, above a transition temperature T_c, we have a macroscopic occupation of the ground state.

We compute this occupation number by bounding $\langle N \rangle$ using the maximum value

$$\langle N \rangle \leq \frac{\zeta(3)}{(\beta\hbar\omega)^3} = \zeta(3) \left(\frac{k_B T}{\hbar\omega}\right)^3.$$

Thus, for a fixed total particle number N_{tot}, we find that the critical temperature is

$$T_c = \left(\frac{\hbar\omega}{k_B}\right)\left(\frac{N_{\text{tot}}}{\zeta(3)}\right)^{1/3}.$$

Above this temperature, the occupation is microscopic and follows the distribution derived above. Below this temperature, the ground state is occupied microscopically; the number of particles in the excited states is limited to our bounding value, and all particles not there are in the ground state.

(f) Below the transition temperature, the number of particles in excited states remains at

$$N = \frac{\zeta(3)}{(\beta\hbar\omega)^3}.$$

The number of particles in the ground state is h ence

$$N_0 = N_{\text{tot}} - \frac{\zeta(3)}{(\beta\hbar\omega)^3}.$$

If we define $f = N_0/N_{\text{tot}}$, we get

$$f = 1 - \frac{\zeta(3)}{N_{\text{tot}}(\beta\hbar\omega)^3}.$$

(g) Below the transition temperature, the particles in the ground state do not contribute any energy, and hence the energy can be computed from the grand canonical partition function approximated using an integral

$$\begin{aligned} E &= -\left(\frac{\partial J}{\partial T}\right) \\ &= \frac{\partial}{\partial T}\left[\frac{(k_B T)^4}{(\hbar\omega)^3}\text{Li}_4(z)\right] \\ &= 4k_B\left(\frac{k_B T}{\hbar\omega}\right)^3 \text{Li}_4(z) \\ &= 4k_B\left(\frac{k_B T}{\hbar\omega}\right)^3 \zeta(4) \end{aligned}$$

where we note that the chemical potential is 0 under the transition temperature. The heat capacity is given by

$$c_V = \frac{\partial E}{\partial T} = 12k_B^2\frac{(k_B T)^2}{(\hbar\omega)^3}\zeta(4). \qquad \square$$

Problem 5. Prove that the entropy per photon in black body radiation is independent of temperature and in d spatial dimensions is given by

$$s = \frac{S}{N} = k_B(d+1)\frac{\sum_{n=1}^{\infty} n^{-d-1}}{\sum_{n=1}^{\infty} n^{-d}} = k_B(d+1)\frac{\zeta(d+1)}{\zeta(d)},$$

where $\zeta(s)$ is the Riemann Zeta function.

Proof. This system is rather similar to that of the harmonic oscillator in the previous problem. In a one-dimensional box of length L, the wavelength is restricted by

$$\lambda = \frac{2L}{n},\ n \in \mathbb{N}.$$

Thuus, the frequency is limited by

$$f = \frac{nc}{2L}$$

and the energy is quantised

$$E = hf = \frac{hc}{2L}n.$$

In d dimensions, the energy is just the sum of the energy in all 3 modes to give

$$E = \frac{hc}{2L}\left(\sum_{i=1}^{d} n_i\right).$$

This is rather reminiscent of the previous problem; we define $\Omega(n)$ analogously. We note that $\Omega(n)$ solves a Stars and Bars problem. Essentially, we imagine arranging the n books in a line and partition them into the k boxes by placing $k-1$ dividers such that the books within a single group into a single box. We can arrange them by imagining the n books and $k-1$ dividers as $n+k-1$ symbols, and we must choose $k-1$ to be dividers. Filling in the rest of the symbols with books yields our partitions. Thus, we have

$$\Omega(n) = \binom{n+d-1}{d-1} = \frac{(n+d-1)!}{(d-1)!n!}.$$

Importantly, we note that for high n this is well approximated by

$$\Omega(n) \approx \frac{1}{(d-1)!}n^{d-1}.$$

Substituting this in the expression from the previous problem, we find that the partition function Z_G is given by

$$\ln Z_G = \sum_{n=0}^{\infty} \Omega(n) \ln\left(1 - ze^{-\beta\frac{hc}{2L}n}\right)$$

$$
\begin{aligned}
&= \int_0^\infty \frac{1}{(d-1)!} n^{d-1} \ln\left(1 - z e^{-\beta \frac{hc}{2L} n}\right) \mathrm{d}n \\
&= \frac{1}{(d-1)!} \frac{1}{\left(\beta \frac{hc}{2L}\right)^d} \int_0^\infty \left(\beta \frac{hc}{2L} n\right)^{d-1} \ln\left(1 - z e^{-\beta \frac{hc}{2L} n}\right) \mathrm{d}\left(\beta \frac{hc}{2L} n\right)
\end{aligned}
$$

We then evaluate the second integral by integration by parts analogously to the previous problem

$$
\begin{aligned}
&\int_0^\infty x^{d-1} \ln\left(1 - z e^{-x}\right) \mathrm{d}x \\
&= \left[\frac{x^d}{d} \ln\left(1 - z e^{-x}\right)\right]_0^\infty - \int_0^\infty \frac{x^d}{d} \frac{1}{1 - z e^{-x}} (z e^{-x}) \, \mathrm{d}x \\
&= \frac{1}{d} \int_0^\infty \frac{x^d}{e^x / z - 1} \, \mathrm{d}x \\
&= \frac{1}{d} \Gamma(d+1) \mathrm{Li}_{d+1}(z)
\end{aligned}
$$

Substituting this yields the grand canonical partition function $J = -k_B T \ln Z_G$

$$
J = -\frac{1}{(d-1)!} \frac{1}{\left(\frac{hc}{2L}\right)^d} (k_B T)^{d+1} \frac{1}{d} \Gamma(d+1) \mathrm{Li}_{d+1}(z).
$$

From here, we can compute the entropy as

$$
\begin{aligned}
S &= -\left(\frac{\partial J}{\partial T}\right)_{V,\mu} \\
&= \frac{d+1}{(d-1)!} \frac{k_B}{\left(\frac{hc}{2L}\right)^d} (k_B T)^d \frac{1}{d} \Gamma(d+1) \mathrm{Li}_{d+1}(z) \\
&= \frac{d+1}{(d-1)!} \frac{k_B}{\left(\frac{hc}{2L}\right)^d} (k_B T)^d \frac{1}{d} \Gamma(d+1) \mathrm{Li}_{d+1}(1)
\end{aligned}
$$

and the expected number of particles as

$$
\begin{aligned}
n &= -\left(\frac{\partial J}{\partial n}\right)_{T,\mu} \\
&= \frac{1}{(d-1)!} \frac{1}{\left(\frac{hc}{2L}\right)^d} (k_B T)^{d+1} \frac{1}{d} \Gamma(d+1) \mathrm{Li}'_{d+1}(z) \\
&= \frac{1}{(d-1)!} \frac{1}{\left(\frac{hc}{2L}\right)^d} (k_B T)^{d+1} \frac{1}{d} \Gamma(d+1) \frac{\mathrm{Li}_d(z)}{z} \frac{\partial z}{\partial \mu}
\end{aligned}
$$

$$= \frac{1}{(d-1)!}\frac{1}{\left(\frac{hc}{2L}\right)^d}(k_BT)^{d+1}\frac{1}{d}\Gamma(d+1)\frac{\mathrm{Li}_d(z)}{z}\beta e^{\beta\mu}$$
$$= \frac{1}{(d-1)!}\frac{1}{\left(\frac{hc}{2L}\right)^d}(k_BT)^{d}\frac{1}{d}\Gamma(d+1)\mathrm{Li}_d(1)$$

We note that in the last line of all of these computations, we have used the fact that photons can be created and destroyed and thus $\mu = 0$ (or $z = 1$) for the photons. Dividing yields

$$\frac{S}{n} = (d+1)k_B\frac{\mathrm{Li}_{d+1}(1)}{\mathrm{Li}_d(1)} = (d+1)k_B\frac{\zeta(d+1)}{\zeta(d)}. \qquad \square$$

Problem 6. Consider two reservoirs for electrons at thermal equilibrium. Refer to them as left L and right R leads which extend in the x direction. Imagine we apply small voltages to the two leads which tunes their chemical potential by eV_L and eV_R.

- Starting from the scattering matrix approach, relate the states entering and leaving each lead at the interface.
- Within the second quantization formalism, write the formula for the electrical current. **Hint:** it will be useful later to write it as an energy integration.
- Taking advantage of the elements of the scattering matrix, identify in the previous formula the transmission function.
- Identify the equilibrium conductance

$$G = \frac{e^2}{h}\,\mathrm{Tr}\left[\mathbf{t}^\dagger(E_F)\,\mathbf{t}(E_F)\right], \tag{1}$$

  where $\mathbf{t}(E_F)$ is the transmission matrix of our system evaluated at the Fermi energy E_F. What is the meaning of the eigenvalues of the hermitian matrix $\mathbf{t}^\dagger(E_F)\mathbf{t}(E_F)$?

Proof. (a) It is necessary before answering this question to do a bit of set up work. We take our sample to extend in the z direction. The leads, on the left and right, have some eigenmodes that are plane waves in the z direction, i.e. they are proportional to $e^{\pm ikz}$. In the transverse direction, we also have some spatial wavefunctions. The total energy is given by the sum of the energies associated with each direction,

$E = E_n + E_t$. The exact structure of this is not important[1]. What is important is that there $E_n \geq 0$, and thus for each value of the total energy E there is a finite number $n(E)$ of modes for this energy in each.

First, we write down the definition of the S-matrix.

$$\begin{pmatrix} \vec{\mathbf{a}}_L^- \\ \vec{\mathbf{a}}_R^+ \end{pmatrix} = \underbrace{\begin{pmatrix} \mathbf{r} & \mathbf{t}' \\ \mathbf{t} & \mathbf{r}' \end{pmatrix}}_{S} \begin{pmatrix} \vec{\mathbf{a}}_L^+ \\ \vec{\mathbf{a}}_R^- \end{pmatrix}.$$

It is important to clarify the definitions of these coefficients $\vec{\mathbf{a}}_{L/R}^{\pm}$. The S-matrix is defined to relate <u>probability currents</u>, and not wavefunction amplitudes. Thus, it is important that we include a prefactor of $k^{-1/2}$ in front of the wavefunctions, i.e. that the wavefunction is formed by (up to constants)

$$\psi \propto a_{L,1}^- \frac{1}{\sqrt{k_1}} e^{-ik_1 z} + \cdots + a_{L,n_L}^- \frac{1}{\sqrt{k_{n_L}}} e^{-ik_{n_L} z} + \text{right movers.}$$

Due to the scattering matrix constraint, the Hilbert space dimension is $2n$ instead of the $4n$ that we would get from $4n$ independent modes. We write creation & annihilation operators $\hat{a}_{L/R,k}^\dagger$, $\hat{a}_{L/R,k}$ for incoming modes in the left & right leads in channel k, and $\hat{b}_{L/R,k}^\dagger$, $\hat{b}_{L/R,k}$ for the incoming modes.

It is important to note here that $\hat{a}_{L,k}^\dagger$ creates an <u>eigenstate</u> with energy E such that there is a probability current of 1 in the k channel in the left lead, and no incoming probability currents in the other channels, or in the right lead. Thus, $\hat{a}_{L,k}^\dagger |0\rangle$ has nonzero incoming probability amplitudes in the outgoing waves. In this language, the annihilation operators are related by

$$\begin{pmatrix} \hat{b}_{L,1} \\ \vdots \\ \hat{b}_{L,n_L} \\ \hat{b}_{R,1} \\ \vdots \\ \hat{b}_{R,n_R} \end{pmatrix} = \begin{pmatrix} \mathbf{r} & \mathbf{t}' \\ \mathbf{t} & \mathbf{r}' \end{pmatrix} \begin{pmatrix} \hat{a}_{L,1} \\ \vdots \\ \hat{a}_{L,n_L} \\ \hat{a}_{R,1} \\ \vdots \\ \hat{a}_{R,n_R} \end{pmatrix}.$$

[1] In principle, it is possible to take the lead to be a cube, from which we would get the transverse modes by solving the 2-dimensional particle in a box, as in the next problem. However, the point here is that it does not matter what shape it actually is, and all this information is hidden away in the S-matrix.

(b) In general, the electrical current can be determined through the derivative $j^\mu = \frac{\delta S[A]}{\delta A^\mu}$, and leads to the expression for free particles coupled to an electric field

$$\begin{aligned}\vec{\mathbf{J}} &= \vec{\mathbf{J}}^{\boldsymbol{\nabla}} + \vec{\mathbf{J}}^{\vec{\mathbf{A}}} \\ \vec{\mathbf{J}}^{\boldsymbol{\nabla}} &= \frac{\hbar}{2mi}\left[\psi^\dagger \boldsymbol{\nabla}\psi - (\boldsymbol{\nabla}\psi^\dagger)\psi\right] \\ \vec{\mathbf{J}}^{\vec{\mathbf{A}}} &= -\frac{q}{m}\vec{\mathbf{A}}\psi^\dagger\psi\end{aligned}$$

In the case of a DC current, we usually have an electric field but no magnetic field. Hence, we can set $\vec{\mathbf{A}} = 0$. We can write this as the current operator

$$\mathbf{J} = \frac{\hbar}{2mi}\left[\vec{\partial} + \overleftarrow{\partial}\right]$$

where the very dumb arrow denotes that the derivative acts to the left or to the right, because we have to write it as a matrix before second quantising it :((

To rewrite this as a field operator, we can promote the wavefunction ψ into an operator valued measure (field), and integrate the current density to obtain the current[2]:

$$\hat{I}_L(z,t) = \frac{\hbar e}{2mi}\int\left[\hat{\Psi}_L^\dagger(\vec{\mathbf{r}},t)\frac{\partial}{\partial z}\hat{\Psi}_L(\vec{\mathbf{r}},t) - \left(\frac{\partial}{\partial z}\hat{\Psi}_L^\dagger(\vec{\mathbf{r}},t)\right)\hat{\Psi}_L(\vec{\mathbf{r}},t)\right]\mathrm{d}\vec{\mathbf{r}}_\perp\,.$$

We would like to write our algebra in terms of the creation and annihilation operators for the energy eigenbasis. We know that the change of basis formula in second quantisation is

$$\hat{\tilde{a}}_\mu = \sum_\nu \langle\tilde{\psi}_\mu|\psi_\nu\rangle\,\hat{a}_\nu,$$

and thus the field operators are given by

$$\begin{aligned}\hat{\Psi}_L(\vec{\mathbf{r}},t) &= \int e^{-iEt/\hbar}\sum_{m=1}^{n_L(E)}\frac{\chi_{L,m}(\vec{\mathbf{r}}_\perp)}{\sqrt{2\pi\hbar v_{L,m}(E)}}\left[\hat{a}_{L,m}e^{ik_{L,m}z} + \hat{b}_{L,m}e^{-ik_{L,m}z}\right]\mathrm{d}E \\ \hat{\Psi}_L^\dagger(\vec{\mathbf{r}},t) &= \int e^{iEt/\hbar}\sum_{m=1}^{n_L(E)}\frac{\chi_{L,m}^*(\vec{\mathbf{r}}_\perp)}{\sqrt{2\pi\hbar v_{L,m}(E)}}\left[\hat{a}_{L,m}^\dagger e^{-ik_{L,m}z} + \hat{b}_{L,m}^\dagger e^{ik_{L,m}z}\right]\mathrm{d}E\,.\end{aligned}$$

Here, $\chi_{L,k}$ is the perpendicular part of the wavefunction, the prefactor of $e^{-iEt/\hbar}$ comes from the time evolution of an energy eigenstate, and $v_{L,k}(E) = \hbar k_{L,m}/m$

[2] Here, we only evaluate the current in the left lead. However, the current in both leads is identical, as can be proven using the unitarity of the S-matrix

is the velocity. We note that the funny factor down there reproduces the desired normalisation.

We are now ready to substitute this back into the expression for the current. However, it is necessary to have a bit of foresight first. Later, we will make the assumption that the leads are thermalised. Thus, the correlation functions will be that of the pure thermal system

$$\langle \hat{b}^\dagger_{\lambda,\eta} \hat{b}_{\lambda',\eta'} \rangle = \delta_{\eta\eta'} \delta_{\lambda\lambda'} f_{\mu_\eta}(E_\lambda).$$

The function f_μ is the Fermi function

$$f_\mu(x) = \frac{1}{e^{\beta(x-\mu)} + 1}$$

and the chemical potentials are

$$\mu_{L/R} = eV_{L/R}.$$

Thus, when expanding, it is not necessary to keep cross terms. This is important as the $\mathrm{d}\vec{\mathbf{r}}_\perp$ integral is not a full scalar product and is not guaranteed to kill all cross terms, and thus we would in principle need to keep all terms in. Thus, we have

$$\hat{\Psi}^\dagger \frac{\partial \hat{\Psi}}{\partial z} = \frac{1}{h} \int \mathrm{d}E \sum_{m=1}^{n_L(E)} \frac{i|\chi_{L,m}|^2 k_{L,m}}{v_{L,m}(E)} (\hat{a}^\dagger_{L,m} \hat{a}_{L,m} - \hat{b}^\dagger_{L,m} \hat{b}_{L,m}) + \text{cross terms}$$

$$\left(\frac{\partial \hat{\Psi}^\dagger}{\partial z} \right) \hat{\Psi} = -\frac{1}{h} \int \mathrm{d}E \sum_{m=1}^{n_L(E)} \frac{i|\chi_{L,m}|^2 k_{L,m}}{v_{L,m}(E)} (\hat{a}^\dagger_{L,m} \hat{a}_{L,m} - \hat{b}^\dagger_{L,m} \hat{b}_{L,m}) + \text{cross terms}$$

Substituting this in the formula for the current and integrating out $\iint |\chi_{L,m}(\vec{\mathbf{r}}_\perp)|^2 \, \mathrm{d}\vec{\mathbf{r}}_\perp = 1$, we get[3]

$$\hat{I}_L(t) = \frac{e}{h} \int \mathrm{d}E \sum_{m=1}^{n_L} \left[\hat{a}^\dagger_{L,m} \hat{a}_{L,m} - \hat{b}^\dagger_{L,m} \hat{b}_{L,m} \right]$$

$$\hat{b}_{L,☺} = \sum_{☹=1}^{n_L} \mathbf{r}_{☺☹} \hat{a}_{L,☹} + \sum_{♪=1}^{n_R} \mathbf{t}'_{☺♪} \hat{a}_{R,♪}$$

$$\hat{I}_L(t) = \frac{e}{h} \int \mathrm{d}E \sum_{m=1}^{n_L} \left[\hat{a}^\dagger_{L,m} \hat{a}_{L,m} - \left(\sum_{☺=1}^{n_L} \mathbf{r}^*_{m☺} \hat{a}^\dagger_{L,☺} + \sum_{☹=1}^{n_R} (\mathbf{t}')^*_{m☹} \hat{a}^\dagger_{R,☹} \right) \right.$$

[3] There were too many sums and I ran out of symbols.

$$\left(\sum_{\text{♪}=1}^{n_L} \mathbf{r}_{m\text{♪}}\hat{a}_{L,\text{♪}} + \sum_{\text{♩}=1}^{n_R} (\mathbf{t}')_{m\text{♩}}\hat{a}_{R,\text{♩}}\right)\Bigg]$$

$$= \frac{e}{h}\int \mathrm{d}E \sum_{m=1}^{n_L}\left[\hat{a}^\dagger_{L,m}\hat{a}_{L,m} - \sum_{☺=1}^{n_L} |\mathbf{r}_{m☺}|^2 \hat{a}^\dagger_{L,☺}\hat{a}_{L,☺} - \sum_{☹=1}^{n_L} |\mathbf{t}'_{m☹}|^2 \hat{a}^\dagger_{R,☹}\hat{a}_{R,☹}\right]$$

Now, it remains to do some massaging. Noting that we will replace $\hat{a}^\dagger_{L/R,☺}\hat{a}_{L/R,☺}$ with the Fermi function after taking the expectation value since these states all have the same energy later, we compute

$$\begin{aligned}\sum_{☺,☹=1}^{n_L} |\mathbf{t}'_{☺☹}|^2 &= \sum_{☺,☹=1}^{n_L} (\mathbf{t}')^\dagger_{☹☺}\mathbf{t}_{☺☹} \\ &= \mathrm{Tr}\Big((\mathbf{t}')^\dagger\mathbf{t}'\Big)\end{aligned}$$

We then note that $\mathrm{Tr}\big((\mathbf{t}')^\dagger\mathbf{t}'\big) = \mathrm{Tr}\big(\mathbf{t}^\dagger\mathbf{t}\big)$ due to the unitarity of the S-matrix:

$$SS^\dagger = \begin{pmatrix}\mathbf{r} & \mathbf{t}' \\ \mathbf{t} & \mathbf{r}'\end{pmatrix}\begin{pmatrix}\mathbf{r}^\dagger & \mathbf{t}^\dagger \\ \mathbf{t}'^\dagger & \mathbf{r}'^\dagger\end{pmatrix} = \begin{pmatrix}\mathbf{r}\mathbf{r}^\dagger + \mathbf{t}'\mathbf{t}'^\dagger & \mathbf{r}\mathbf{t}^\dagger + \mathbf{t}'\mathbf{r}'^\dagger \\ \mathbf{t}\mathbf{r}^\dagger + \mathbf{r}'\mathbf{t}'^\dagger & \mathbf{t}\mathbf{t}^\dagger + \mathbf{r}'\mathbf{r}'^\dagger\end{pmatrix} = \mathbb{1}$$

$$S^\dagger S = \begin{pmatrix}\mathbf{r}^\dagger & \mathbf{t}^\dagger \\ \mathbf{t}'^\dagger & \mathbf{r}'^\dagger\end{pmatrix}\begin{pmatrix}\mathbf{r} & \mathbf{t}' \\ \mathbf{t} & \mathbf{r}'\end{pmatrix} = \begin{pmatrix}\mathbf{r}^\dagger\mathbf{r} + \mathbf{t}^\dagger\mathbf{t} & \mathbf{r}^\dagger\mathbf{t}' + \mathbf{t}^\dagger\mathbf{r}' \\ \mathbf{t}'^\dagger\mathbf{r} + \mathbf{r}'^\dagger\mathbf{t} & \mathbf{t}'^\dagger\mathbf{t}' + \mathbf{r}'^\dagger\mathbf{r}'\end{pmatrix} = \mathbb{1}$$

Looking at the upper left diagonal component, which is the identity matrix, and taking its transpose, we find that

$$\mathbf{t}'\mathbf{t}'^\dagger = \mathbf{t}^\dagger\mathbf{t}$$

and hence using the properties of the trace we get

$$\mathrm{Tr}\Big(\mathbf{t}'^\dagger\mathbf{t}'\Big) = \mathrm{Tr}\Big(\mathbf{t}^\dagger\mathbf{t}\Big).$$

We do the same for the the other term. Nearing the end, we must look at the top left component of the equation $S^\dagger S = \mathbb{1}$ equation again to find

$$\mathbb{1} - \mathbf{r}^\dagger\mathbf{r} = \mathbf{t}^\dagger\mathbf{t}$$

and hence

$$n_L - \mathrm{Tr}\Big(\mathbf{r}^\dagger\mathbf{r}\Big) = \mathrm{Tr}\Big(\mathbf{t}^\dagger\mathbf{t}\Big).$$

Thus, after taking the expectation value, we get

$$I_L = \langle \hat{I}_L(t)\rangle = \int \frac{e}{h}\,\mathrm{Tr}\Big(\mathbf{t}^\dagger\mathbf{t}\Big)\,[f_L(E) - f_R(E)]\,\mathrm{d}E\,.$$

(c) The transmission function is the trace expression in the previous equation.

(d) The conductance can be evaluated at 0 temperature and for small voltage differences as follows: At small temperatures, the fermi function is a step function. f_L goes to 0 at $E_F + eV_L$, while f_R goes to 0 at $E_F + eV_R$. If V_L is higher than V_R, then f_L goes to 0 earlier than f_R. This is the only part where $f_L - f_R$ does not vanish, and it has width $e(V_L - V_R) =: eV$, with V the potential difference. Additionally, if the voltages are small, we can take the integrand to be constant and evaluate it at E_F. Thus,

$$I_L = \frac{e^2 V}{h} \operatorname{Tr}\left(\mathbf{t}^\dagger \mathbf{t}\right).$$

Identifying this with the classical ohmic expression

$$I_L = VG,$$

we arrive at the expression

$$G = \frac{e^2}{h} \operatorname{Tr}\left(\mathbf{t}^\dagger \mathbf{t}\right). \qquad \square$$

Problem 7. Consider a wide conductor along x, e.g. width in y direction is large W while the height in z direction is small. The length in the x direction is large L.

Given information: The density of states of a 2D spin degenerate system is

$$\mathcal{D}_0 = \frac{m}{\pi \hbar^2}.$$

The transmission through a wire with length L is

$$T = \frac{L_0}{L + L_0},$$

where L_0 is the mean free path. The conductivity is

$$\sigma = e^2 \mathcal{D}_0 D,$$

where

$$D = \frac{v_F L_0}{\pi}$$

is the diffusion coefficient and v_F is the Fermi velocity.

Using this information, relate the previous results for conductance G with Ohm's law. I.e., does the relationship between G and conductivity σ correspond to Ohm's law?

Proof. We begin by recalling the results for the eigenenergies of a particle in a 1-dimensional box:

$$E_n = \frac{\hbar^2 k_n^2}{2m}, \qquad k_n = \frac{n\pi}{L}.$$

Since the height in the z direction is very small, the k values are very high and spaced very far apart. Thus, we can assume that we are only in the lowest mode in the z direction.

Now, we want to count the number of conducting modes. The total Hamiltonian is

$$H = \frac{1}{2m}(p_x^2 + p_y^2).$$

We seek to choose wavenumbers k_x, k_y such that

$$k_x^2 + k_y^2 = k_F^2.$$

Since the length in x is very large, for each k_y, we should be able to pick a k_x such that $k_x^2 + k_y^2 \approx k_F^2$. Thus, we simply seek to count the k_y such that

$$|k_y| \leq |k_F|$$

or

$$\frac{n\pi}{W} \leq |k_F|.$$

Thus, the total number of modes is

$$N = \frac{k_F W}{\pi}.$$

The total conductivity is given by (initial 2 from spin degeneracy)

$$\begin{aligned}
G &= \frac{2e^2}{h}\left(\frac{k_F W}{\pi}\right)\underbrace{\frac{L_0}{L+L_0}}_{T} \\
&= \frac{2e^2}{h}\frac{W}{\pi}\frac{mv_F}{\hbar}\frac{L_0}{L+L_0} \\
&= \frac{e^2 W m v_F}{\pi^2\hbar^2}\frac{L_0}{L+L_0} \\
&\approx \frac{e^2 W m v_F}{\pi^2\hbar^2}\frac{L_0}{L} \\
&= \left(\frac{e^2 L_0 m v_F}{\pi^2\hbar^2}\right)\frac{W}{L}
\end{aligned}$$

We have used the diffusive limit $L \gg L_0$ here. From Ohm's law, we identify the term in parentheses as the conductivity. We can also evaluate the given expression

$$\begin{aligned}\sigma &= e^2 \mathcal{D}\mathcal{D} \\ &= e^2 \left(\frac{m}{\pi\hbar^2}\right)\left(\frac{v_F L_0}{\pi}\right) \\ &= \frac{e^2 L_0 m v_F^2}{\pi^2 \hbar^2}\end{aligned}$$

This agrees with the conductivity derived earlier from the total conductance and Ohm's law. Thus, we conclude that our result is consistent with Ohm's law. □

The following exercises are intended to understand the London–Pippard phenomenological theory. We are going to derive them with a simple model. For that we start considering the supercurrent can be thought as a nonviscous charged fluid with field velocity $\vec{v}(\vec{x}, t)$, that obeys

$$\vec{j}(\vec{x}, t) = -n_s e \vec{v}(\vec{x}, t),$$

where n_s is the density of the super-electron number density and $-e$ is the electronic charge. The continuity equation together with Newton's second law give

$$\nabla \cdot \vec{j} = \nabla \cdot \vec{v} = 0,$$

$$\frac{d\vec{v}}{dt} = -\frac{e}{m}\left(\vec{E} + \frac{1}{c}\vec{v} \times \vec{h}\right),$$

with $\vec{h}$ the microscopic field which is larger compared to the atomic size but smaller compared to the penetration depth. Notice that this also involves the total derivative with respect to time of the velocity and not just its partial (explicit) dependence.

Problem 8.

- Define the effective field

$$\vec{Q} = \nabla \times \vec{v} - \frac{e}{mc}\vec{h}.$$

  Together with Maxwell's equations, derive the dynamical equation for $\vec{v}$ for a superconducting bulk under the constraint $\vec{Q} = 0$. **Hint:** Notice that in Newton's second law equation it was written d/dt and not $\partial/\partial t$. Recall for a field vector which is the relation between those two and use it to get the equation $\partial\vec{v}/\partial t$.

- How can we interpret $\vec{Q}$?

Proof. First, a comment on the units: Due to the factor of $1/c$ in the Lorentz force, the equation is written in Gaussian units. We also note that $\vec{h}$ is the "B-field", i.e. it does not have a prefactor related to the permeability already absorbed in. In these units, the Maxwell's equations are given by

$$\begin{aligned}
\nabla \cdot \vec{E} &= 4\pi\rho \\
\nabla \cdot \vec{B} &= 0 \\
\nabla \times \vec{E} + \frac{1}{c}\frac{\partial \vec{B}}{\partial t} &= 0 \\
\nabla \times \vec{B} - \frac{1}{c}\frac{\partial \vec{E}}{\partial t} &= \frac{4\pi}{c}\vec{j}
\end{aligned}$$

The material derivative is given by

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{\partial f}{\partial t} + \vec{v} \cdot \nabla f.$$

We apply this to $f = \vec{v}$ to get

$$\begin{aligned}
\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} &= -\frac{e}{m}\left(\vec{E} + \frac{1}{c}\vec{v} \times \vec{h}\right) \\
&= -\frac{e}{m}\vec{E} - \vec{v} \times (\nabla \times \vec{v})
\end{aligned}$$

We recall a certain lemma, which can be found on lists of vector calculus identities

Lemma 1.

$$\vec{v} \cdot \nabla \vec{v} = \nabla\left(\frac{v^2}{2}\right) - \vec{v} \times (\nabla \times \vec{v}).$$

This leads to

$$\frac{\partial \vec{v}}{\partial t} = \nabla\left(\frac{v^2}{2}\right) - \frac{e}{m}\vec{E}.$$

The vector $\vec{Q}$ can be interpreted as a generalised vorticity

$$\begin{aligned}
\vec{Q} &= \nabla \times \vec{v} - \frac{e}{mc}\vec{h} \\
&= \nabla \times \vec{v} - \frac{e}{mc}\nabla \times \vec{A} \\
&= \nabla \times \left(\vec{v} - \frac{e}{mc}\vec{A}\right).
\end{aligned}$$

□

Problem 9. Now consider a semi-infinite slab in the 3D space that occupies $z > 0$. One of the equations of the previous item can be rewritten as

$$\vec{h} = -\frac{mc}{n_s e^2}\nabla \times \vec{j}.$$

- Rewrite the previous equation using the Maxwell equations for static fields to write $\vec{h}$ in terms of spatial derivatives of $\vec{h}$.

- Consider an applied field $\vec{H} = H_0\hat{x}$ parallel to the surface. Recalling that inside the superconductor the field has to be zero, find an acceptable solution for the microscopic field. What is the penetration depth? Identify the London penetration depth λ_L.

- If the slab has a $2d$ thickness, under the same external field, can you find an appropriate solution for $\vec{h}(z)$? Check that it is a solution of the differential equation that you found before. Consider the mean magnetic field in the superconductor as the spatial average of $h(z)$. What can you say about the Meissner effect when $d \ll \lambda_L$ and $d \gg \lambda_L$?

Proof.

- We have the Maxwell's equation

$$\boldsymbol{\nabla} \times h = \frac{4\pi}{c}\vec{j}.$$

Then, substituting this in, we get

$$\begin{aligned}\vec{h} &= -\frac{mc}{n_s e^2}\boldsymbol{\nabla} \times \left(\frac{c}{4\pi}\boldsymbol{\nabla} \times \vec{h}\right) \\ &= -\frac{mc^2}{4\text{ø}\pi n_s e^2}(\boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \vec{h}) - \nabla^2\vec{h}) \\ &= \frac{mc^2}{4\pi n_s e^2}\nabla^2\vec{h}\end{aligned}$$

where we also used the equation $\boldsymbol{\nabla} \cdot \vec{h} = 0$.

- We define the penetration depth

$$\lambda_L = \sqrt{\frac{mc^2}{4\pi n_s e^2}}.$$

With this definition, the equation is written as

$$\vec{h} = \lambda_L^2 \nabla^2 \vec{h}.$$

In one-dimension, this is

$$\vec{h} = \lambda_L^2 \frac{\partial^2 \vec{h}}{\partial z^2}.$$

Which has solution

$$\vec{h} = H_0 \hat{x} e^{-z/\lambda_L}.$$

- We are looking for a solution in one dimension to the differntial equation

$$f(z) = \lambda_T^2 \frac{\partial^2 f}{\partial z^2}$$

with boundary conditions

$$f(0) = f(2d) = 1$$

representing that the magnetic field extends indefinitely outside the superconductor. This is a simple second order simple harmonic oscillator like equation. As typical, we substitute in the ansatz $f = Ae^{kx}$ to get the equation

$$k^2 \lambda_L^2 = 1$$

which has solutions $k = \pm 1/\lambda_L$. Thus, our solution is of the form

$$f = Ae^{z/\lambda_L} + Be^{-z/\lambda_L}.$$

We then substitute in the boundary conditions to get

$$\begin{aligned} A + B &= 1 \\ Ae^{2d/\lambda_L} + Be^{-2d/\lambda_L} &= 1 \end{aligned}$$

This has the solution

$$A = \frac{1}{1 + e^{2d/\lambda_L}}, \qquad B = \frac{1}{1 + e^{-2d/\lambda_L}}.$$

Thus, the general solution is

$$\vec{h} = H_0 \hat{x} \left(\frac{e^{z/\lambda_L}}{1 + e^{2d/\lambda_L}} + \frac{e^{-z/\lambda_L}}{1 + e^{-2d/\lambda_L}} \right).$$

In the limit of $d \gg \lambda_L$, we consider two limits: When $z \ll d$, the exponential term with z in it remains bounded and does not diverge in any way. Thus, we can approximate

$$\frac{1}{1 + e^{2d/\lambda_L}} \approx 0, \qquad \frac{1}{1 + e^{-2d/\lambda_L}} \approx 1$$

to get as solution in this limit

$$\vec{h} = H_0 \hat{x} e^{-z/\lambda_L}, \qquad z \ll d.$$

In the limit $z \approx d$, we get

$$\frac{e^{-z/\lambda_L}}{1+e^{-2d/\lambda_L}} \approx 0.$$

as the denominator stays greater than 1, while the numerator tends to 0 as $z \gg \lambda_L$. On the other side, we approximate

$$\frac{e^{z/\lambda_L}}{1+e^{2d/\lambda_L}} \approx e^{(z-2d)/\lambda_L}$$

which is just the exponentially decaying solution, but flipped. Thus, in the $d \gg \lambda_L$ limit, the solution is just two exponentially decaying solutions glued together. In the limit $d \ll \lambda_L$, both denominators approximate to 2, and we have

$$\begin{aligned}\vec{h} &\approx \frac{H_0\hat{x}}{2}\left(e^{z/\lambda_L}+e^{-z/\lambda_L}\right)\\ &\approx \frac{H_0\hat{x}}{2}(2)\\ &= H_0\hat{x}\end{aligned}$$

showing that the magnetic field has not been expelled from the superconductor. □

Problem 10. Now we can find a conservation law. For simplicity we start with linearized equations:

$$-\frac{1}{n_s e}\frac{\partial \vec{j}}{\partial t} = \frac{\partial \vec{v}}{\partial t} = -\frac{e}{m}\vec{E}.$$

Consider a surface S bounded by a fixed closed curve $\mathcal{C}$ that lies wholly in the superconducting material. S can be wherever placed. Integrating Maxwell's equations one has

$$\int_S d\vec{S}\cdot\frac{\partial \vec{h}}{\partial t} = -c\int_S d\vec{S}\cdot(\nabla\times\vec{E}) = -c\oint_{\mathcal{C}} d\vec{\ell}\cdot\vec{E}.$$

- Find the conserved quantity.
- Can you add more assumptions to get more specific results? For instance:
  - Suppose that your curve $\mathcal{C}$ is far from the boundaries.
  - Suppose the interior of $\mathcal{C}$ is wholly superconducting.
- It is interesting to rewrite the conserved quantity in terms of the vector potential $\vec{A}$ and momentum $\vec{p}$. Retrieve the conserved quantity. What does it remind you of?

Proof.

- We begin by substituting from the first equation to get

$$\begin{aligned}\int_S \mathrm{d}\vec{S} \cdot \frac{\partial \vec{h}}{\partial t} &= -c \oint_C \mathrm{d}\vec{\ell} \cdot \vec{E} \\ &= \frac{mc}{e} \oint_C \mathrm{d}\vec{\ell} \cdot \frac{\partial \vec{v}}{\partial t}\end{aligned}$$

which leads to the equation

$$\frac{\partial}{\partial t}\left(\int_S \mathrm{d}\vec{S} \cdot \vec{h} + \frac{mc}{e} \oint_C \mathrm{d}\vec{\ell} \cdot \vec{v}\right) = 0.$$

This tells us that the quantity

$$\int_S \mathrm{d}\vec{S} \cdot \vec{h} + \frac{mc}{e} \oint_C \mathrm{d}\vec{\ell} \cdot \vec{v}$$

is conserved.

We can replace this with local conserved quantities by using Stoke's theorem on either of the two quantities. Since $\vec{h} = \boldsymbol{\nabla} \times \vec{A}$, we have

$$\frac{\partial}{\partial t} \oint \left(\vec{A} + \frac{mc}{e}\vec{v}\right) \cdot \mathrm{d}\vec{l} = 0.$$

Since this holds for all contours C, we find that the quantity in the integral is locally conserved, and this is (up to constants) the quantity in the definition of the vector $\vec{Q}$.

We can also do this the other way around to get

$$\frac{\partial}{\partial t} \int_S \left(\vec{h} + \frac{mc}{e}\boldsymbol{\nabla} \times v\right) \cdot \mathrm{d}S$$

and again the quantity in parentheses is conserved.

- If the interior of C is wholly superconducting, the magnetic field inside is 0 due to the Meissner effect, and thus we find that the vorticity is conserved:

$$\frac{\partial}{\partial t}\left(\boldsymbol{\nabla} \times \vec{v}\right) = 0.$$

Note that it is in principle possible for S to be anywhere - even outside the superconductor, even when the interior of C is inside. However, because $\boldsymbol{\nabla} \cdot h = 0$, this does not matter, and the argument still works. □

Problem 11. Choosing the London gauge $\nabla \cdot \vec{A} = 0$ and $\vec{A} \cdot \hat{n}$ ($\hat{n}$ surface normal) for the vector potential we have

$$\vec{j} = -\frac{n_s e^2}{mc}\vec{A}.$$

We are now going to consider a generalization of this equation where $\vec{j}$ is determined as a spatial average of $\vec{A}$ throughout some neighboring region of dimension r_0. The motivation behind this is to take into account non-local superconducting effects. In heavily doped alloys, r_0 is comparable with the electronic mean free path l in the normal metal. We introduce the Pippard coherence length ξ_0:

$$\frac{1}{r_0} = \frac{1}{\xi_0} + \frac{1}{l}.$$

Pippard then rewrote the London equation as an average over space for the field $\vec{A}$:

$$\vec{j}(\vec{x}) = -\frac{n_s e^2}{mc}\frac{3}{4\pi\xi_0}\int d^3\vec{X}\,\frac{(\vec{X}\cdot\vec{A}(\vec{x}'))}{X^4}e^{-X/r_0},$$

where $\vec{X} = \vec{x} - \vec{x}'$.

- That equation has to be solved together with its corresponding Maxwell equation, making it quite cumbersome to solve. Nevertheless we can extract physical features. Consider $r_0 \ll \lambda_L$ with λ_L the London penetration length and get a relation for $\vec{j}$ and $\vec{A}$.

- The previous result was obtained under the assumption $r_0 \ll \lambda_L$. Consider the other limit $r_0 \gg \lambda_L$ and consider a current sheet $j_0\delta(z)\hat{y}$ in the plane. The current sheet generates a magnetic field which then induces a supercurrent $\vec{j}$. Write the Maxwell equation for $\vec{h}$ that has to be solved together with the Pippard equation.

  - Use Fourier transformations to find the (linear) equation that relates $\vec{j}$ to $\vec{A}$ in the momentum space.
  - Write the final result for $\vec{A}$ and $\vec{h}$. Note that it is enough to have them expressed as integrals.
  - Can you find the proper limit to recover the Meissner effect?

Proof.

- Since $r_0 \ll \lambda_L$, we can approximate $\vec{A}(\vec{x}') = \vec{A}(\vec{x})$. Then in each component we have to evaluate

$$\int \mathrm{d}^3x'\,\frac{\vec{x}'(\vec{x}'\cdot\vec{A})}{(|\vec{x}'|)^4}e^{-|\vec{x}'|/r_0}.$$

First, we note that the off diagonal integrals of the form

$$\int \mathrm{d}^3x' \, \frac{x'y'}{(|\vec{x}'|)^4} e^{-|\vec{x}'|/r_0}$$

vanish because they are odd in x' or y'. Hence we evaluate the diagonal integrals

$$\int \mathrm{d}^3x' \, \frac{(x')^2}{(|\vec{x}'|)^4} e^{-|\vec{x}'|/r_0}.$$

We convert to spherical coordinates and note that all of the cartesian coordinates x', y' or z' are proportional to r^2. Then, including the jacobian, the integral converts to

$$\int \mathrm{d}\Omega \int_0^\infty \mathrm{d}r \, e^{-r/r_0} f(\Omega) \, \mathrm{d}r \, .$$

The r integral yields r_0. Then, after performing the Ω integral, we find that

$$\vec{j}(\vec{x}) = -\frac{n_s e^2}{mc} \frac{r_0}{\xi_0}.$$

Finally, since $r_0 \ll l$, we find that $r_0 \approx \xi_0$ and recover the equation

$$\vec{j} = -\frac{n_s e^2}{mc} \vec{A}.$$

- We begin with the Maxwell's equation (in Gaussian units, where we have already used the Couloumb gauge)

$$\nabla^2 \vec{A} = -\frac{4\pi}{c} \vec{j}.$$

We perform a Fourier transform on Pippard's equation, using the convolution theorem

$$\int_{-\infty}^{\infty} f(\tau) g(t-\tau) \, \mathrm{d}\tau = \int_{-\infty}^{\infty} f(t-\tau) g(\tau) \, \mathrm{d}\tau = \hat{f}(\omega) \hat{g}(\omega).$$

By performing a Fourier transformation on the convolution kernel, we get

$$\vec{j}(\vec{q}) = -\frac{c}{4\pi} K(\vec{q}) \vec{A}(\vec{q})$$

with

$$K(\vec{q}) = \frac{3}{2\lambda_L^2} \frac{1}{qr_0} \left[\frac{\tan^{-1}(qr_0)}{(qr_0)^2} (1 + (qr_0)^2) - \frac{1}{qr_0} \right].$$

At the same time, the Maxwell's equation simplifies to

$$|\vec{q}|^2 \vec{A}(\vec{q}) = -\frac{4\pi}{c} \vec{j}(\vec{q}).$$

Noting that we can split the current up into $\vec{j} = \vec{j}_{\text{supercurrent}} + \vec{j}_{\text{external}}$, we get

$$q^2 \vec{A}(\vec{q}) = \frac{4\pi}{c}\left(j_0 \hat{y} - \frac{c}{4\pi} K(\vec{q}) \vec{A}(\vec{q})\right)$$

which solves to

$$\vec{A}(\vec{q}) = \frac{4\pi}{c} \frac{j_0 \hat{y}}{q^2 + K(\vec{q})}.$$

We find the vector potential in position space by performing a fourier inversion

$$\vec{A}(\vec{x}) = \frac{2\hat{y}}{c} \int \frac{j_0}{|\vec{q}|^2 + K(\vec{q})} e^{i\vec{q}\cdot\vec{x}} \, \mathrm{d}^3\vec{q}\,.$$

The magnetic field $\vec{h}$ is given by the curl of $\vec{A}$:

$$\vec{h} = \boldsymbol{\nabla} \times \vec{A} = \frac{2}{c} \int \frac{j_0(-q_z \hat{x} + q_x \hat{z})}{|\vec{q}|^2 + K(\vec{q})} e^{i\vec{q}\cdot\vec{x}} \, \mathrm{d}^3\vec{q}\,.$$

- The limit that we seek is $qr_0 \ll 1$. In this limit, we can expand $\tan^{-1}(x)$ to find that

  $$K(\vec{q}) \to \frac{1}{\lambda_L^2}$$

  and find that the vector potential decays exponentially due to the mass term: The fourier transform of $\exp(-a|x|)$ is given by

  $$\mathcal{F}\left(e^{-a|x|}\right) = \frac{2a}{a^2 + \omega^2}.$$

  Thus, we see that the magnetic field also decays exponentially, which is the Meissner effect. □

Problem 12. Show that the Ginzburg-Landau free energy of a domain wall can be written as

$$\Delta F = A \frac{u}{4} \int dx \left[\psi_0^4 - \psi^4(x)\right],$$

where $A = L^{d-1}$ is the area of the domain wall. Using this result, show that the surface tension $\sigma = \Delta F / A$ is given by

$$\sigma = \frac{\sqrt{8}}{3} \xi u \psi_0^4,$$

with ξ the correlation length. To get the final result, you will need to recall the functional form of the soliton, i.e., the solution to the Ginzburg-Landau equation through the domain wall.

Proof. Landau theory consists of expanding the free energy as a function of the order parameter near to the phase transition, when the free energy is small:

$$F_L(\psi) = \int \frac{r}{2}\psi^2 + \frac{u}{4}\psi^4 - h\psi \,\mathrm{d}V\,,$$

where h describes a conjugate field such as a magnetic field. To describe the possible effects of a spatially varying order parameter, we include a gradient term

$$F_{GL}(\psi) = \int \frac{s}{2}(\boldsymbol{\nabla}\psi)^2 + \frac{r}{2}\psi^2 + \frac{u}{4}\psi^4 - h\psi \,\mathrm{d}V\,.$$

We seek solutions, i.e. order parameter fields that minimise the Ginzburg-Landau functional. This is done by functional differentiation to obtain

$$\begin{aligned}
\delta F_{GL}(\psi) &= \int s(\boldsymbol{\nabla}\psi)\cdot\boldsymbol{\nabla}\delta\psi + r\psi\delta\psi + u\psi^3\delta\psi - h\delta\psi \,\mathrm{d}V \\
&= \int s\boldsymbol{\nabla}\cdot(\delta\psi\boldsymbol{\nabla}\psi) - s(\nabla^2\psi)\delta\psi + r\psi\delta\psi + u\psi^3\delta\psi - h\delta\psi \,\mathrm{d}V \\
&= \int -s(\nabla^2\psi)\delta\psi + r\psi\delta\psi + u\psi^3\delta\psi - h\delta\psi \,\mathrm{d}V
\end{aligned}$$

This leads to the equation

$$-s(\nabla^2\psi) + r\psi + u\psi^3 - h = 0.$$

The domain wall we seek is a solition like solution where the order parameter goes from one minimum to the other within a finite range of x. For $h = 0$ (no external field), these minima are symmetric, and we will call them ψ_0. Defining the derivative of the Landau free energy as

$$f_L'[\psi] = r\psi + u\psi^3,$$

we can summarise by saying that a domain wall is a solution of the equation

$$s\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = f_L'[\psi(x)]$$

with asymptotic boundary conditions

$$\psi \xrightarrow{x\to\infty} = \psi_0, \qquad \psi \xrightarrow{x\to-\infty} = -\psi_0.$$

Due to the symmetry in all directions other than x, the free energy is the Ginzburg-Landau energy multiplied by the area A of the other dimensions., we find that

$$F[\psi] = A\int \frac{s}{2}(\boldsymbol{\nabla}\psi)^2 + \frac{r}{2}\psi^2 + \frac{u}{4}\psi^4 \,\mathrm{d}V$$

$$
\begin{aligned}
&= A \int -\frac{s}{2}\psi\nabla^2\psi + \frac{r}{2}\psi^2 + \frac{u}{4}\psi^4 \,\mathrm{d}V \\
&= A \int -\frac{\psi}{2}\left(r\psi + u\psi^3\right) + \frac{r}{2}\psi^2 + \frac{u}{4}\psi^4 \,\mathrm{d}V \\
&= -A \int \frac{u}{4}\psi^4 \,\mathrm{d}V
\end{aligned}
$$

We need to subtract off the energy associated with a uniform configuration ψ_0, which is clearly given by

$$F_0[\psi_0] = A \int \frac{r}{2}\psi_0^2 + \frac{u}{4}\psi_0^4 \,\mathrm{d}V\,.$$

Since ψ_0 must minimise the Landau free energy, we must have

$$r\psi_0 + u\psi_0^3 = 0$$

which leads to

$$F_0[\psi_0] = A \int -\frac{u}{2}\psi_0^4 + \frac{u}{4}\psi_0^4 \,\mathrm{d}V = -A \int \frac{u}{4}\psi_0^4.$$

This, of course, could also have been derived by substituting ψ_0 for ψ in the earlier expression for $F[\psi]$. Subtracting, we find that the energy of the domain wall is given by

$$\Delta F = A\frac{u}{4} \int \mathrm{d}x\, [\psi_0^4 - \psi^4(x)].$$

Before we proceed further, we will need to solve the equation for the order parameter field explicitly. We can solve this by analogy to Newton's Second Law with f_L' as the potential. Defining $v = \frac{\mathrm{d}\psi}{\mathrm{d}x}$, we can rewrite the equation using the chain rule as

$$sv\frac{\mathrm{d}v}{\mathrm{d}\psi} = f_L'[\psi] = r\psi + u\psi^3.$$

We note that the initial conditions are that at the start, $v = 0$ and $\psi = -\psi_0$. Thus, separating variables yields

$$
\begin{aligned}
\int_0^v sv\,\mathrm{d}v &= \int_{-\psi_0}^{\psi} r\psi + u\psi^3 \,\mathrm{d}\psi \\
\frac{1}{2}sv^2 &= \left[\frac{r}{2}\psi^2 + \frac{u}{4}\psi^4\right]_{-\psi_0}^{\psi} \\
&= \frac{r}{2}(\psi^2 - \psi_0^2) + \frac{u}{4}(\psi^4 - \psi_0^4)
\end{aligned}
$$

Again, we simplify this by noting that ψ_0 was the minimum of the Landau energy and hence satisfied the equation

$$r + u\psi_0^2 = 0.$$

Substituting, we find that

$$\begin{aligned}\frac{1}{2}sv^2 &= -\frac{u\psi_0^2}{2}(\psi^2-\psi_0^2)+\frac{u}{4}(\psi^4-\psi_0^4)\\ &= u\left[-\frac{1}{2}\psi^2\psi_0^2+\frac{1}{2}\psi_0^4+\frac{1}{4}\psi^4-\frac{1}{4}\psi_0^4\right]\\ &= \frac{u}{4}\left[\psi_0^4+\psi^4-2\psi^2\psi_0^2\right]\\ &= \frac{u}{4}(\psi_0^2-\psi^2)^2\end{aligned}$$

Since $v=\psi'$ must always be of the same sign, we choose the positive sign to allow us to travel from $-\psi_0$ to ψ_0. Thus, this simplifies to

$$\begin{aligned}v &= \sqrt{\frac{u}{2s}}(\psi_0^2-\psi^2)\\ &= \frac{\psi_0}{\sqrt{2}\xi}\left(1-\frac{\psi^2}{\psi_0^2}\right)\end{aligned}$$

where ξ is defined as the correlation length $\xi=\sqrt{s/|r|}$. We can then integrate this differential equation to get

$$\frac{\psi_0}{\sqrt{2}\xi}\,\mathrm{d}x=\frac{1}{1-\frac{\psi^2}{\psi_0^2}}\,\mathrm{d}\xi\,.$$

This yields the desired solution

$$\psi(x)=\psi_0\tanh\left(\frac{x-x_0}{\sqrt{2}\xi}\right).$$

Now all we need to do is substitute this back in to get

$$\begin{aligned}\sigma=\frac{\Delta F}{A} &= \frac{u}{4}\psi_0^4\int_{-\infty}^{\infty}\left[1-\tanh^4\left(\frac{x-x_0}{\sqrt{2}\xi}\right)\right]\mathrm{d}x\\ &= \frac{\xi u}{\sqrt{8}}\psi_0^4\int_{-\infty}^{\infty}\left[1-\tanh^4(u)\right]\mathrm{d}u\\ &= \frac{\sqrt{8}}{3}\xi u\psi_0^4.\end{aligned}$$

□

Problem 13. Consider a two-component Dirac electron moving in one dimension through a domain wall, described by the wave equation

$$\left(-i\sigma_1\nabla_x-m(x)\sigma_3\right)\psi=E\psi,$$

where the mass field forms a domain wall, changing sign at the origin according to

$$m(x)=m_0\tanh\left(\frac{x}{\sqrt{2}\xi}\right).$$

Asymptotically, the energy of the excitation is gapped, with an excitation spectrum $E(k) = \sqrt{k^2 + m_0^2}$. Show that the domain wall gives rise to a zero energy bound state and derive the form of its wave function.

Proof. Since it is given that the bound state has 0 energy, we seek such states directly:

$$(-i\sigma_1 \nabla_x - m(x)\sigma_3)\psi = 0.$$

This corresponds to, in terms of the components, if $\psi = (\psi_1, \psi_2)^T$

$$\begin{pmatrix} i\partial_x \psi_2 + m(x)\psi_1 \\ i\partial_x \psi_1 + m(x)\psi_2 \end{pmatrix} = 0.$$

Alternatively, we could also write

$$\frac{\partial}{\partial x}\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = -im(x)\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = m(x)\sigma_2 \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}$$

This is a linear differential equation and can be solved using the matrix exponential

$$\begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix} = \exp\left(\sigma_2 \int_0^x m(x)\,\mathrm{d}x\right)\begin{pmatrix} \psi_1(0) \\ \psi_2(0) \end{pmatrix}$$

While this matrix exponential can be explicitly evaluated, we can proceed in a more clever manner. The zero energy state must be normalisable, and hence must decay at $\pm\infty$. Since the function $m(x)$ is even, we can write

$$\begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix} = \exp\left(\sigma_2 \int_0^{|x|} m(x)\,\mathrm{d}x\right)\begin{pmatrix} \psi_1(0) \\ \psi_2(0) \end{pmatrix}$$

from which it is clear that the integral is always positive. Now, since σ_2 has eigenvalues ± 1, we must choose the negative eigenvector $(1, -i)$. Thus, we find that the state is given by, up to a constant prefactor,

$$\begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix} = \exp\left(\sigma_2 \int_0^{|x|} m(x)\,\mathrm{d}x\right)\begin{pmatrix} 1 \\ -i \end{pmatrix} = \exp\left(-\int_0^{|x|} m(x)\,\mathrm{d}x\right)\begin{pmatrix} 1 \\ -i \end{pmatrix}. \qquad \square$$

Problem 14. Let us consider an infinite multimoded wire, i.e. a 2D system that is infinite in the x direction but finite in the y direction. The Hamiltonian for the wire is

$$H = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + U(y). \tag{1}$$

Here $U(y)$ is the confining potential in the y direction. Derive the Green's function for a semi-infinite wire using eigenfunction expansion (Lehmann representation),

$$G^r(\mathbf{x}, \mathbf{x}') = -\frac{i}{\hbar v_m} \sum_m \chi_m(y)\chi_m(y') e^{ik_m|x-x'|}, \tag{2}$$

with

$$k_m = \sqrt{\frac{2m}{\hbar^2}(E - \varepsilon_{m,0})}, \qquad v_m = \frac{\hbar k_m}{m},$$

and $\chi_m(y)$ the transverse eigenfunctions.

Proof. The two particle retarded correlation function is given by

$$C^+(t) = -i\theta(t)\langle[\hat{X}_1(t), \hat{X}_2(0)]_\zeta\rangle.$$

Derivation of the Lehmann Representation

We are interested in the two particle retarded Green's function, which is given by

$$C^+(t) = -i\theta(t)\left(\langle\hat{X}_1(t)\hat{X}_2(0)\rangle - \zeta\langle\hat{X}_2(0)\hat{X}_1(t)\rangle\right)$$

where the thermal expectation value is given by

$$\langle\dots\rangle = \frac{1}{Z}\operatorname{Tr}\left(\dots e^{-\beta(\hat{H}-\mu\hat{N})}\right).$$

We define the eigenstates of $\hat{H} - \mu\hat{N}$ as $|\alpha\rangle$ with eigenvalue E_α. Then, we can compute the two point correlation function

$$\begin{aligned}
\langle\hat{X}_1(t)\hat{X}_2(0)\rangle &= \frac{1}{Z}\sum_\alpha \langle\alpha|\left(\hat{X}_1(t)\hat{X}_2(0)\right) e^{-\beta(\hat{H}-\mu\hat{N})}|\alpha\rangle \\
&= \frac{1}{Z}\sum_\alpha \langle\alpha|\left(\hat{X}_1(t)\hat{X}_2(0)\right)|\alpha\rangle\, e^{-\beta E_\alpha} \\
&= \frac{1}{Z}\sum_{\alpha,\beta} \langle\alpha|\hat{X}_1(t)|\beta\rangle\ \langle\beta|\hat{X}_2(0)|\alpha\rangle\, e^{-\beta E_\alpha} \\
&= \frac{1}{Z}\sum_{\alpha,\beta} X_{1,\alpha\beta}(t) X_{2,\beta\alpha} e^{-\beta E_\alpha}
\end{aligned}$$

with the matrix elements defined appropriately. Next, we note that the time dependence comes from the Heisenberg representations to compute

$$\begin{aligned}
X_{1,\alpha\beta}(t) &= \langle\alpha|e^{it(\hat{H}-\mu\hat{N})} X_1 e^{-it(\hat{H}-\mu\hat{N})}|\beta\rangle \\
&= e^{itE_\alpha}\,\langle\alpha|X_1|\beta\rangle\, e^{-itE_\beta}
\end{aligned}$$

$$= e^{it(E_\alpha - E_\beta)} X_{1,\alpha\beta}$$

We then perform an analogous computation on the other correlation function to get

$$\begin{aligned}
\langle \hat{X}_2(0)\hat{X}_1(t)\rangle &= \frac{1}{Z}\sum_{\alpha,\beta} \zeta \, \langle \alpha|\hat{X}_2(0)|\beta\rangle \, \langle \beta|\hat{X}_1(t)|\alpha\rangle \, e^{-\beta E_\alpha} \\
&= \frac{1}{Z}\sum_{\alpha,\beta} X_{2,\alpha\beta} X_{1,\beta\alpha} e^{-it(E_\alpha - E_\beta)} e^{-\beta E_\alpha} \\
&= \frac{1}{Z}\sum_{\alpha,\beta} X_{2,\beta\alpha} X_{1,\alpha\beta} e^{-it(E_\beta - E_\alpha)} e^{-\beta E_\beta}
\end{aligned}$$

Thus, we find that the correlation function is equal to

$$C^+(t) = -\frac{i\theta(t)}{Z}\sum_{\alpha,\beta} X_{1,\alpha\beta} X_{2,\beta\alpha} e^{it(E_\alpha - E_\beta)} \left[e^{-\beta E_\alpha} - \zeta e^{-\beta E_\beta}\right].$$

This can be fourier transformed into frequency space to give us

$$C^+(\omega) = \frac{1}{Z}\sum_{\alpha\beta} X_{1\alpha\beta} X_{2\beta\alpha} \left[\frac{e^{-\beta E_\alpha}}{\omega + E_{\alpha\beta} + i\eta} + \frac{e^{-\beta E_\beta}}{\omega + E_{\alpha\beta} + i\eta}\right]$$

where $E_{\alpha\beta} = E_\alpha - E_\beta$.

In this case, we are looking for the field operators $X_1 = \Psi(\vec{\mathbf{x}}),\ X_2 = \Psi^\dagger(\vec{\mathbf{x}}')$. The field operators are defined in terms of the single body eigenstates ϕ_n by

$$\Psi(\vec{\mathbf{x}}) = \sum_n \phi_n(\vec{\mathbf{x}}) c_n, \qquad \Psi(\vec{\mathbf{x}}') = \sum_n \phi_n^*(\vec{\mathbf{x}}') c_n^\dagger.$$

Substituting this in, we find that

$$\begin{aligned}
C^+(\omega) &= \frac{1}{Z}\sum_{\alpha\beta n n'} \phi_n(\vec{\mathbf{x}})\phi_{n'}^*(\vec{\mathbf{x}}') \, \langle \alpha|c_n|\beta\rangle \, \langle \beta|c_{n'}^\dagger|\alpha\rangle \left[\frac{e^{-\beta E_\alpha}}{\omega + E_{\alpha\beta} + i\eta} + \frac{e^{-\beta E_\beta}}{\omega + E_{\alpha\beta} + i\eta}\right] \\
&= \frac{1}{Z}\sum_{\alpha\beta n} \phi_n(\vec{\mathbf{x}})\phi_n^*(\vec{\mathbf{x}}') \, \langle \alpha|c_n|\beta\rangle \, \langle \beta|c_n^\dagger|\alpha\rangle \left[\frac{e^{-\beta E_\alpha}}{\omega + E_{\alpha\beta} + i\eta} + \frac{e^{-\beta E_\beta}}{\omega + E_{\alpha\beta} + i\eta}\right]
\end{aligned}$$

We note that this means that state β must have an additional mode as compared to state α created by $c_n^\dagger$. Thus, we have $E_{\alpha\beta} = -\epsilon_n$, and $e^{-\beta E_\alpha} + e^{-\beta E_\beta} = e^{-\beta E_\alpha}(1 + e^{-\beta\epsilon_n})$. Thus,

the nonvanishing terms simplify to

$$C^+(\omega) = \frac{1}{Z}\sum_{n\alpha} e^{-\beta E_\alpha}\phi_n(\vec{\mathbf{x}})\phi_n^*(\vec{\mathbf{x}}')\frac{1+e^{-\beta\epsilon_n}}{\omega-\epsilon_n+i\eta}.$$

Before we move on, we must talk about the thermal factor. We note that the only α that contribute are those where the orbital n are empty. Then, we split the sum to find

$$C^+(\omega) = \sum_n \phi_n(\vec{\mathbf{x}})\phi_n^*(\vec{\mathbf{x}}')\frac{1}{\omega-\epsilon_n+i\eta}\frac{1}{Z}\sum_\alpha\left[e^{-\beta E_\alpha}+e^{-\beta(E_\alpha+\epsilon_n)}\right]$$

We see that the two terms in the sum provide exactly the sum over all possible many body states that appear in the partition function. Hence, they cancel out and we find that

$$C^+(\omega) = \sum_n \phi_n(\vec{\mathbf{x}})\phi_n^*(\vec{\mathbf{x}}')\frac{1}{\omega-\epsilon_n+i\eta}.$$

Now it is finally time to make use of the structure of the problem. Since the Hamiltonian can be separated into two parts concerning the x and y directions, n is actually a multiindex consisting of (k,m). The wavefunctions are given by

$$\phi_{m,k} = \frac{1}{\sqrt{2\pi}}e^{ikx}\chi_m(y), \qquad \epsilon_{m,k} = \epsilon_{m,0}+\frac{\hbar^2k^2}{2m}.$$

We substitute this back and replace the sum over states with the measure $\sum_m \int \mathrm{d}k$ to get

$$\begin{aligned}
C^+(E) &= \sum_m \int \frac{\chi_m(y)\chi_m^*(y')e^{ik(x-x')}}{E-\epsilon_{m,0}-\frac{\hbar^2k^2}{2m}+i\eta}\frac{\mathrm{d}k}{2\pi} \\
&= \frac{1}{2}\sum_m \chi_m(y)\chi_m(y')\left[\frac{\exp\left(i\sqrt{\frac{2m}{\hbar^2}(E-\epsilon_{m,0}+i\eta)}|x-x'|\right)}{\frac{\hbar^2}{2m}\sqrt{\frac{2m}{\hbar^2}\left(E-\epsilon_{m,0}+i\eta\right)}}\right] \\
&= -\frac{i}{\hbar}\sum_m \frac{1}{v_m}\chi_m(y)\chi_m(y')e^{ik_m|x-x'|}.
\end{aligned}$$

□

Problem 15. The scattering matrix is related to the Green function via the Fisher-Lee relation

$$s_{nm} = -\delta_{nm} + \frac{i\hbar\sqrt{v_n v_m}}{a}\int dy_p dy_q\, \chi_n(y_q)G^r(y_q,y_p)\chi_m(y_p), \tag{3}$$

where $G^r(y_q,y_p)$ is the retarded Green function connecting a point in lead p to one in lead q.

Derive the expression for the transmission probability from lead q to lead p,

$$T_{pq} = \mathrm{Tr}\left[\Gamma_p G^r \Gamma_q G^a\right], \tag{4}$$

and compare with the result from the Normal Metals exercise sheet.

Proof. First we note that there is actually no δ because the two relate to different leads. In fact, as written, assuming p and q are different leads, this is only the $\mathbf{t}$ part of the scattering matrix. However, it is all we need. We compute

$$\begin{aligned}
&\mathrm{Tr}\left(\mathbf{t}^\dagger \mathbf{t}\right) \\
=&\sum_{mn} t^*_{mn} t_{mn} \\
=&\frac{\hbar^2}{a^2}\sum_{mn} v_n v_m \int \mathrm{d}y_p\,\mathrm{d}y_q\,\mathrm{d}y'_p\,\mathrm{d}y'_q\,\chi_n(y_q)\chi^*_n(y'_q)G^r(y_q,y_p)G^a(y'_q,y'_p)\chi_m(y_p)\chi^*_m(y'_p)
\end{aligned}$$

where we note that complex conjugation turns G^r into G^a by shifting the poles from slightly above the real axis to slightly below ($i\eta \to -i\eta$). To turn this into a matrix product form, we write

$$\Gamma_q(y_q,y'_q) = \frac{\hbar}{a}\sum_n v_n\chi_n(y_q)\chi^*_n(y_q), \qquad \Gamma_p(y_p,y'_p) = \frac{\hbar}{a}\sum_m v_m\chi_m(y_p)\chi^*_m(y'_p)$$

and find that

$$\mathrm{Tr}\left(\mathbf{t}^\dagger \mathbf{t}\right) = \int \Gamma_p(y_p,y'_p)G^r(y_q,y_p)\Gamma_q(y_q,y'_q)G^a(y'_q,y'_p)\,\mathrm{d}y_p\,\mathrm{d}y_q\,\mathrm{d}y'_p\,\mathrm{d}y'_q$$

which is (up to a possible reordering of the function arguments) in matrix product form as shown above. Using the identification of the Fisher-Lee relation as the elements of the transmission matrix, we can see that this is equivalent to the result in the normal metals sheet. □

Problem 16 (Phonon-mediated electron–electron interaction)**.** Consider the following Hamiltonian

$$H = H_{\text{el-gas}} + H_{\text{ph}} + \sum_{kq} g_q\left(a_q + a^\dagger_{-q}\right)c^\dagger_{k+q,\sigma}c_{k\sigma},$$

where $H_{\text{el-gas}}$ describes a free electronic gas, and H_{ph} describes the free phonon part. The last term describes an interaction between the electrons and phonons with coupling constant g_q, and where $c_{k\sigma}$ (a_q) describes an electronic (phononic) destruction operator.

Since the phonon operators appear in the combination $a_q + a^\dagger_{-q}$, it is often convenient to define the phonon Green's function in terms of the combined operator

$$A_q = a_q + a^\dagger_{-q}.$$

(a) Find the bare phonon Green's function from the definition

$$D_0(q,t) = -i\langle\Phi_0|T\left[A_q(t)A_q^\dagger(0)\right]|\Phi_0\rangle. \tag{6}$$

Here $|\Phi_0\rangle$ is the phonon vacuum/ground state and T is the time-ordering operator. Use the interaction picture where

$$A_q(t) = e^{iH_0t}A_qe^{-iH_0t}, \qquad H_0 = H_{\text{ph}} = \sum_q \omega_q a_q^\dagger a_q.$$

(b) Apply a Fourier transform to frequency space to get

$$D_0(q,\omega) = \frac{2\omega_q}{\omega^2 - \omega_q^2 + i\eta},$$

where $\eta = 0^+$. <u>Note: during the Fourier transform you will need to introduce factors $i\delta$, with $\delta = 0^+$, as appropriate to make sure the integrals converge.</u>

(c) Draw a Feynman diagram depicting an effective electron–electron interaction mediated by the phonon propagator D_0, i.e. glue two phonon lines together from the diagram depicting electron–phonon coupling. Also assume the two incoming electrons have opposite momentum $k, -k$ and opposite spin.

From this, read off the interaction strength (with $k' = k + q$) as

$$V_{k,k'} = |g_{k'-k}|^2\, D_0(k'-k,\omega).$$

We often take the static limit $\omega = 0$ and assume the phonon-mediated attraction is only active for electrons with an energy smaller than the Debye frequency (maximum phonon energy) relative to the Fermi level. Show that the interaction becomes independent of momentum if we take the simple jellium model acoustic phonons

$$g_q = g\sqrt{q}, \qquad \omega_q = uq.$$

Proof. (a) We can write the time ordered product as

$$T[A_q(t)A_q^\dagger(0)] = \theta(t)A_q(t)A_q^\dagger(0) + \zeta\theta(-t)A_q^\dagger(0)A_q(t)$$

where $\zeta = 1$ due to the bosonic statistics. Then we compute each expectation value separately:

$$\langle A_q(t)A_q^\dagger(0)\rangle = \langle\Phi_0|\left(e^{iH_0t}A_qe^{-iH_0t}A_q^\dagger\right)|\Phi_0\rangle$$

$$
\begin{aligned}
&= \langle\Phi_0|\left(A_q e^{-iH_0 t} A_q^\dagger\right)|\Phi_0\rangle \\
&= \langle\Phi_0|\left(a_q e^{-iH_0 t} a_{-q}^\dagger\right)|\Phi_0\rangle \\
&= \langle\Phi_0|\left(a_q e^{-i\omega_q t} a_q^\dagger\right)|\Phi_0\rangle \\
&= e^{-i\omega_q t}
\end{aligned}
$$

and

$$
\begin{aligned}
\langle A_q^\dagger(0) A_q(t)\rangle &= \langle\Phi_0|\left(A_q^\dagger e^{iH_0 t} A_q e^{-iH_0 t}\right)|\Phi_0\rangle \\
&= \langle\Phi_0|\left(A_q^\dagger e^{iH_0 t} A_q\right)|\Phi_0\rangle \\
&= \langle\Phi_0|\left(a_{-q} e^{iH_0 t} a_{-q}^\dagger\right)|\Phi_0\rangle \\
&= e^{i\omega_{-q} t}
\end{aligned}
$$

Noting that the phonon dispersion is even due to the symmetry of the lattice, we find that

$$D_0(q,t) = -i[\theta(t)e^{-i\omega_q t} + \theta(-t)e^{i\omega_q t}].$$

(b) We use the convention

$$\mathcal{F}(f)(\omega) = \hat{f}(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}\,\mathrm{d}t\,.$$

We start with the first term as an example and compute

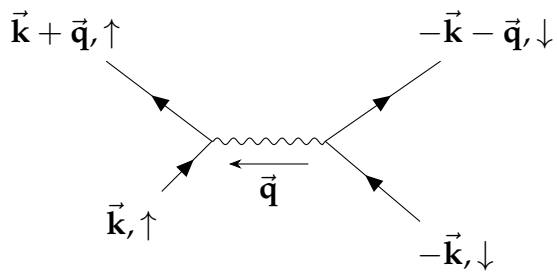
$$
\begin{aligned}
\mathcal{F}\left(\theta(t)e^{-i\omega_q t}\right) &= \int_{-\infty}^{\infty} \theta(t)e^{-i(\omega+\omega_q)t}e^{-\eta t}\,\mathrm{d}t \\
&= \int_{0}^{\infty} e^{-i(\omega+\omega_q)t}e^{-\eta t}\,\mathrm{d}t \\
&= \frac{1}{\eta + i(\omega+\omega_q)} \\
\mathcal{F}\left(\theta(-t)e^{i\omega_q t}\right) &= \int_{-\infty}^{\infty} \theta(-t)e^{i(\omega_q-\omega)t}e^{\eta t} \\
&= \int_{-\infty}^{0} e^{i(\omega_q-\omega)t}e^{\eta t}\,\mathrm{d}t \\
&= \frac{1}{\eta - i(\omega-\omega_q)}
\end{aligned}
$$

Then, substituting this in, we find that

$$D_0(q,\omega) = -i\left[\frac{1}{\eta + i(\omega+\omega_q)} + \frac{1}{\eta - i(\omega-\omega_q)}\right]$$

$$
\begin{aligned}
&= -i\left[\frac{\eta + i(\omega+\omega_q) + \eta - i(\omega-\omega_q)}{[\eta + i(\omega+\omega_q)][\eta - i(\omega-\omega_q)]}\right] \\
&= \frac{2\omega_q - 2i\eta}{\omega^2 - \omega_q^2 + i\eta} \\
&= \frac{2\omega_q}{\omega^2 - \omega_q^2 + i\eta}
\end{aligned}
$$

(c) Can you suggest a good package for feynman diagrams this looks horrible :((



Clarifying note on what we are doing: If we perform perturbation theory on the electron-phonon interaction, we will find terms like

$$
\langle O \rangle = \int \left[\sum_{kk'qq'} g_q g_{q'} O A_q A_{q'} \overline{\psi}_{k+q,\sigma} \psi_{k,\sigma} \overline{\psi}_{k'+q',\sigma} \psi_{k',\sigma} \right] e^{-S_{\text{el}}[\psi,\overline{\psi}] - S_{\text{ph}}[z,\overline{z}]} \mathcal{D}(\psi,\overline{\psi}) \mathcal{D}(z,\overline{z})
$$

If we want to compute 2 particle correlation functions, i.e. have two creation and two annihilation operators in O, we connect Green's functions into to the vertex elements to get the diagram above. Since the variables A and those of ψ are uncorrelated (corresponding to two different actions), we just find that

$$
\langle O \rangle = \int \left[\sum_{kk'qq'} g_q g_{q'} O \langle A_q A_{q'} \rangle \overline{\psi}_{k+q,\sigma} \psi_{k,\sigma} \overline{\psi}_{k'+q',\sigma} \psi_{k',\sigma} \right] e^{-S_{\text{el}}[\psi,\overline{\psi}]} \mathcal{D}(\psi,\overline{\psi})
$$

By an electron-electron interaction, we mean an interaction vertex that reproduces the correlation functions. While this diagram is a 2nd order perturbation theory diagram, the phonons can in principle be integrated out exactly because the action is Gaussian. If we had a two-particle interaction $S_{\text{int}} = \sum_{0123} V_{0123} \overline{\psi}_3 \psi_1 \overline{\psi}_2 \psi_0$, the expectation values would be given by

$$
\langle O \rangle = \int \left[\sum_{0123} O V_{0123} \overline{\psi}_3 \psi_1 \overline{\psi}_2 \psi_0 \right] e^{-S_{\text{el}}[\psi,\overline{\psi}]} \mathcal{D}(\psi,\overline{\psi})
$$

where 0123 are just shorthand for phase space coordinates. Thus, by comparison, setting $k' = -k$ and noting that $q' = -q$ for the functional average to be nonvanishing, we find that the effective interaction is given by

$$V_{k,k'} = |g_{k'-k}|^2 \, D_0(k' - k, \omega).$$

where the propagator D_0 is the functional average $\langle A_q A_{q'} \rangle$ (note implicit time ordering in path integral).

If we take the jellium model acoustic phonons $g_q = g\sqrt{q}$, $\omega_q = uq$ we find that

$$\begin{aligned} V_{k,k'} &= |g_{k'-k}|^2 D_0(k' - k, \omega) \\ &= g^2 |q| \frac{2\omega_q}{\omega^2 - \omega_q^2 + i\eta} \\ &= g^2 |q| \frac{2u|q|}{\omega^2 - u^2 q^2 + i\eta} \\ &= g^2 |q| \frac{2u|q|}{-u^2 q^2 + i\eta} \\ &= -\frac{2g^2}{u}. \end{aligned}$$

□

Problem 17 (Cooper pairs with finite momentum)**.** Generalize Cooper's calculation to a pair with finite momentum. In particular:

(a) Show that the operator that creates a Cooper pair at a finite momentum $\vec{p}$,

$$\Lambda_{\vec{p}}^{\dagger} = \int d^3x \, d^3x' \, \phi(\vec{x} - \vec{x}') \psi_{\uparrow}^{\dagger}(\vec{x}) \psi_{\downarrow}^{\dagger}(\vec{x}') e^{i\vec{p}\cdot(\vec{x}+\vec{x}')/2}, \tag{9}$$

can be rewritten in the form

$$\Lambda^{\dagger}(\vec{p}) = \sum_{\vec{k}} \phi(\vec{k}) \, c_{\vec{k}+\vec{p}/2,\uparrow}^{\dagger} c_{-\vec{k}+\vec{p}/2,\downarrow}^{\dagger}. \tag{10}$$

(b) Show that the energy E_p of the pair state $\Lambda^{\dagger}(\vec{p})|\mathrm{FS}\rangle$ is given by the (complex) roots $z = E_p$ of the equation

$$1 + \frac{g_0}{V} \sum_{0<\epsilon_k<\omega_D} \frac{1}{z - (\epsilon_{k+p/2} + \epsilon_{k-p/2})} = 0. \tag{11}$$

Here

$$|\mathrm{FS}\rangle = \prod_{k<k_F} c_{k\uparrow}^{\dagger} c_{k\downarrow}^{\dagger} |0\rangle$$

defines the filled Fermi sea, ϵ_k is the energy of an electron with momentum k, g_0 is the strength of the phonon-mediated electron–electron interaction, V is a volume factor, and ω_D is the Debye frequency.

Proof. (a) We begin by performing a fourier transform

$$\psi^\dagger(\vec{\mathbf{r}}) = \frac{1}{\sqrt{V}} \sum_{\vec{\mathbf{k}}} e^{-i\vec{\mathbf{k}}\vec{\mathbf{r}}} c^\dagger_{\vec{\mathbf{k}}} \qquad \psi(\vec{\mathbf{r}}) = \frac{1}{\sqrt{V}} \sum_{\vec{\mathbf{k}}} e^{i\vec{\mathbf{k}}\vec{\mathbf{r}}} c_{\vec{\mathbf{k}}}.$$

Substituting, we find that

$$\begin{aligned}
\Psi^\dagger_{\vec{\mathbf{p}}} &= \int \mathrm{d}^3x\, \mathrm{d}^3x'\, \phi(\vec{\mathbf{x}} - \vec{\mathbf{x}}')\psi^\dagger_\uparrow(\vec{\mathbf{x}})\psi^\dagger_\downarrow(\vec{\mathbf{x}}')e^{i\vec{\mathbf{p}}(\vec{\mathbf{x}}+\vec{\mathbf{x}}')/2} \\
&= \frac{1}{V} \sum_{\vec{\mathbf{k}},\vec{\mathbf{k}}'} \int \mathrm{d}^3x\, \mathrm{d}^3x'\, \phi(\vec{\mathbf{x}} - \vec{\mathbf{x}}')e^{-i\vec{\mathbf{k}}\vec{\mathbf{x}}}e^{-i\vec{\mathbf{k}}'\vec{\mathbf{x}}'} c^\dagger_{\vec{\mathbf{k}},\uparrow} c^\dagger_{\vec{\mathbf{k}}',\downarrow} e^{i\vec{\mathbf{p}}(\vec{\mathbf{x}}+\vec{\mathbf{x}}')/2} \\
&= \frac{1}{V} \sum_{\vec{\mathbf{k}},\vec{\mathbf{k}}'} \int \mathrm{d}^3x\, \mathrm{d}^3x'\, \phi(\vec{\mathbf{x}} - \vec{\mathbf{x}}')c^\dagger_{\vec{\mathbf{k}},\uparrow} c^\dagger_{\vec{\mathbf{k}}',\downarrow} e^{i(-\vec{\mathbf{k}}+\vec{\mathbf{p}}/2)\vec{\mathbf{x}}}e^{i(-\vec{\mathbf{k}}'+\vec{\mathbf{p}}/2)\vec{\mathbf{x}}'} \\
&= \frac{1}{V} \sum_{\vec{\mathbf{k}},\vec{\mathbf{k}}'} \int \mathrm{d}^3R\, \mathrm{d}^3x'\, \phi(\vec{\mathbf{R}})c^\dagger_{\vec{\mathbf{k}},\uparrow} c^\dagger_{\vec{\mathbf{k}}',\downarrow} e^{i(-\vec{\mathbf{k}}+\vec{\mathbf{p}}/2)(\vec{\mathbf{x}}'+\vec{\mathbf{R}})}e^{i(-\vec{\mathbf{k}}'+\vec{\mathbf{p}}/2)\vec{\mathbf{x}}'} \\
&= \frac{1}{V} \sum_{\vec{\mathbf{k}},\vec{\mathbf{k}}'} \int \mathrm{d}^3x'\, \phi(\vec{\mathbf{k}} - \vec{\mathbf{p}}/2)c^\dagger_{\vec{\mathbf{k}},\uparrow} c^\dagger_{\vec{\mathbf{k}}',\downarrow} e^{i(-\vec{\mathbf{k}}+\vec{\mathbf{p}}/2)\vec{\mathbf{x}}'}e^{i(-\vec{\mathbf{k}}'+\vec{\mathbf{p}}/2)\vec{\mathbf{x}}'} \\
&= \sum_{\vec{\mathbf{k}},\vec{\mathbf{k}}'} \phi(\vec{\mathbf{k}} - \vec{\mathbf{p}}/2)c^\dagger_{\vec{\mathbf{k}},\uparrow} c^\dagger_{\vec{\mathbf{k}}',\downarrow} \delta_{\vec{\mathbf{p}}-\vec{\mathbf{k}}-\vec{\mathbf{k}}'} \\
&= \sum_{\vec{\mathbf{k}}} \phi(\vec{\mathbf{k}} - \vec{\mathbf{p}}/2)c^\dagger_{\vec{\mathbf{k}},\uparrow} c^\dagger_{-\vec{\mathbf{k}}+\vec{\mathbf{p}},\downarrow} \\
&= \sum_{\vec{\mathbf{k}}} \phi(\vec{\mathbf{k}})c^\dagger_{\vec{\mathbf{k}}+\vec{\mathbf{p}}/2,\uparrow} c^\dagger_{-\vec{\mathbf{k}}+\vec{\mathbf{p}}/2,\downarrow}
\end{aligned}$$

(b) We write the eigenvalue equation $H\left|\psi\right\rangle = E\left|\psi\right\rangle$ to get

$$E_{\vec{\mathbf{p}}} \sum_{\vec{\mathbf{k}}} \phi_{\vec{\mathbf{k}}} \left|\vec{\mathbf{k}}, \vec{\mathbf{p}}\right\rangle = \sum_{|\vec{\mathbf{k}}+\vec{\mathbf{p}}/2| \geq k_F} (\epsilon_{\vec{\mathbf{k}}+\vec{\mathbf{p}}/2} + \epsilon_{\vec{\mathbf{k}}-\vec{\mathbf{p}}/2})\phi_{\vec{\mathbf{k}}} \left|\vec{\mathbf{k}}, \vec{\mathbf{p}}\right\rangle + \sum_{\vec{\mathbf{k}},\vec{\mathbf{k}}'} \left|\vec{\mathbf{k}}, \vec{\mathbf{p}}\right\rangle \left\langle\vec{\mathbf{k}}, \vec{\mathbf{p}}\right| V \left|\vec{\mathbf{k}}', \vec{\mathbf{p}}\right\rangle \phi_{\vec{\mathbf{k}}',\vec{\mathbf{p}}}$$

where we have defined

$$\left|\vec{\mathbf{k}}, \vec{\mathbf{p}}\right\rangle = c^\dagger_{\vec{\mathbf{k}}+\vec{\mathbf{p}}/2,\uparrow} c^\dagger_{-\vec{\mathbf{k}}+\vec{\mathbf{p}}/2,\downarrow} \left|FS\right\rangle.$$

Here, we note that since V does not change the total momentum, even though $\left|\vec{\mathbf{k}}, \vec{\mathbf{p}}\right\rangle\left\langle\vec{\mathbf{k}}, \vec{\mathbf{p}}\right|$ is not a full resolution of the identity (which would involve a sum over $\vec{\mathbf{p}}$), it still does not change the expression. Comparing coefficients of $\left|\vec{\mathbf{k}}, \vec{\mathbf{p}}\right\rangle$ and

considering the special case where the matrix element is a constant $\left\langle \vec{\mathbf{k}}, \vec{\mathbf{p}} \middle| V \middle| \vec{\mathbf{k}}, \vec{\mathbf{p}} \right\rangle = -g_0/V$ just like in the BCS model, we find the equation

$$E_{\vec{\mathbf{p}}}\phi_{\vec{\mathbf{k}}} = (\epsilon_{\vec{\mathbf{k}}+\vec{\mathbf{p}}/2} - \epsilon_{\vec{\mathbf{k}}-\vec{\mathbf{p}}/2})\phi_{\vec{\mathbf{k}}} - \frac{g_0}{V} \sum_{0<\epsilon_{\vec{\mathbf{k}}\pm\vec{\mathbf{p}}/2}<\omega_D} \phi_{\vec{\mathbf{k}}'}$$

where the interaction is once again limited to pairs with energy less than the Debye frequency. We can solve this to find

$$\phi_{\vec{\mathbf{k}}} = \frac{1}{\epsilon_{\vec{\mathbf{k}}+\vec{\mathbf{p}}/2} + \epsilon_{\vec{\mathbf{k}}-\vec{\mathbf{p}}/2} - E_{\vec{\mathbf{p}}}} \frac{g_0}{V} \sum_{0<\epsilon_{\vec{\mathbf{k}}\pm\vec{\mathbf{p}}/2}<\omega_D} \phi_{\vec{\mathbf{k}}'}.$$

Finally, we sum over $\vec{\mathbf{k}}$ and cancel out the common sum of $\phi_{\vec{\mathbf{k}}'}$ to find

$$1 + \frac{g_0}{V} \sum_{0<\epsilon_{\vec{\mathbf{k}}\pm\vec{\mathbf{p}}/2}<\omega_D} \frac{1}{E_{\vec{\mathbf{p}}} - \epsilon_{\vec{\mathbf{k}}+\vec{\mathbf{p}}/2} - \epsilon_{\vec{\mathbf{k}}-\vec{\mathbf{p}}/2}} = 0.$$

This is the desired equation that $E_{\vec{\mathbf{p}}}$ must solve. □

Problem 18. The BCS wavefunction in momentum space can be written as

$$|\Psi_{\text{BCS}}\rangle = \prod_{\mathbf{k}} \exp\left(\phi_{\mathbf{k}} c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow}\right)|0\rangle = \prod_{\mathbf{k}} \left(1 + \phi_{\mathbf{k}} c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow}\right)|0\rangle. \tag{5}$$

- Show that the BCS ground state is the vacuum for the Bogoliubov quasiparticles, i.e. that the destruction operators $\alpha_{\mathbf{k},\sigma}$ annihilate the BCS ground state. Here

$$\alpha_{\pm\mathbf{k},\sigma} = u_{\mathbf{k}} c_{\pm\mathbf{k}\sigma} + v_{\mathbf{k}} c^\dagger_{\mp\mathbf{k}\bar{\sigma}}, \tag{6}$$

  and you may assume that $u_{\mathbf{k}}, v_{\mathbf{k}}$ are real for convenience.

- If the Bogoliubov quasiparticle is $\alpha^\dagger_{\mathbf{k}\uparrow} = u_{\mathbf{k}} c^\dagger_{\mathbf{k}\uparrow} + v_{\mathbf{k}} c_{-\mathbf{k}\downarrow}$, then starting from the equation of motion of the Bogoliubov quasiparticle

$$[H, \alpha^\dagger_{\mathbf{k}\uparrow}] = \frac{\partial \alpha^\dagger_{\mathbf{k}\uparrow}}{\partial \tau} = E_{\mathbf{k}} \alpha^\dagger_{\mathbf{k}\uparrow}, \tag{7}$$

  show that $(u_{\mathbf{k}}, v_{\mathbf{k}})^T$ is an eigenvector of

$$h_{\mathbf{k}} = \begin{pmatrix} u_{\mathbf{k}} \\ v_{\mathbf{k}} \end{pmatrix} = \begin{pmatrix} \epsilon_{\mathbf{k}} & \Delta \\ \Delta & -\epsilon_{\mathbf{k}} \end{pmatrix} \begin{pmatrix} u_{\mathbf{k}} \\ v_{\mathbf{k}} \end{pmatrix} = E_{\mathbf{k}} \begin{pmatrix} u_{\mathbf{k}} \\ v_{\mathbf{k}} \end{pmatrix} \tag{8}$$

  By solving the eigenvalue problem and assuming the gap is real, give an explicit expression for $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ in terms of Δ and $\epsilon_{\mathbf{k}}$.

Proof. We apply the annihilation operators directly. For concreteness, we will consider $\alpha_{\vec{\mathbf{k}},\uparrow}$. Since we can commute α through all bilinears $1+\phi_{\vec{\mathbf{q}}}c^\dagger_{\vec{\mathbf{q}}}c^\dagger_{\vec{\mathbf{q}}}$ with $\vec{\mathbf{q}} \neq \vec{\mathbf{k}}$, we will ignore them all and simply look at the product

$$(u_{\mathbf{k}}c_{\mathbf{k},\uparrow}+v_{\mathbf{k}}c^\dagger_{-\mathbf{k},\downarrow})(1+\phi_{\vec{\mathbf{k}}}c^\dagger_{\vec{\mathbf{k}},\uparrow}c^\dagger_{-\vec{\mathbf{k}},\downarrow})(1+\phi_{-\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}},\uparrow}c^\dagger_{\vec{\mathbf{k}},\downarrow})$$

applied to the ground state. This product would yield $2^3 = 8$ terms. However, we can immediately see that some fo them drop out. The product $c_{\vec{\mathbf{k}},\uparrow}$ with the two 1s would yield an annihilation operator applied to the vacuum; it vanishes. We may not multiply $c^\dagger_{-\vec{\mathbf{k}},\downarrow}$ with itself; squares of fermion operators vanish. Thus, two more terms drop out. If we multiply $c_{\vec{\mathbf{k}},\uparrow}$ with something not containing $c^\dagger_{\vec{\mathbf{k}},\uparrow}$, we could simply anticommute it to the end, where it would annihilate the ground state again. Thus, there are only 4 terms left, and we find

$$\begin{aligned}
&(u_{\vec{\mathbf{k}}}c_{\vec{\mathbf{k}},\uparrow}+v_{\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}},\downarrow})(1+\phi_{\vec{\mathbf{k}}}c^\dagger_{\vec{\mathbf{k}},\uparrow}c^\dagger_{-\vec{\mathbf{k}},\downarrow})(1+\phi_{-\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}},\uparrow}c^\dagger_{\vec{\mathbf{k}},\downarrow})\\
=\,& u_{\vec{\mathbf{k}}}c_{\vec{\mathbf{k}},\uparrow}\phi_{\vec{\mathbf{k}}}c^\dagger_{\vec{\mathbf{k}},\uparrow}c^\dagger_{-\vec{\mathbf{k}},\downarrow}+u_{\vec{\mathbf{k}}}c_{\vec{\mathbf{k}},\uparrow}\phi_{\vec{\mathbf{k}}}c^\dagger_{\vec{\mathbf{k}},\uparrow}c^\dagger_{-\vec{\mathbf{k}},\downarrow}\phi_{-\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}},\uparrow}c^\dagger_{\vec{\mathbf{k}},\downarrow}\\
&+v_{\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}},\downarrow}+v_{\vec{\mathbf{k}}}\phi_{-\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}},\downarrow}\phi_{-\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}},\uparrow}c^\dagger_{\vec{\mathbf{k}},\downarrow}\\
=\,& -u_{\vec{\mathbf{k}}}\phi_{\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}},\downarrow}-u_{\vec{\mathbf{k}}}\phi_{\vec{\mathbf{k}}}\phi_{-\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}},\downarrow}c^\dagger_{-\vec{\mathbf{k}},\uparrow}c^\dagger_{\vec{\mathbf{k}},\downarrow}\\
&+v_{\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}},\downarrow}+v_{\vec{\mathbf{k}}}\phi_{-\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}},\downarrow}\phi_{-\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}},\uparrow}c^\dagger_{\vec{\mathbf{k}},\downarrow}\\
=\,&\phi_{-\vec{\mathbf{k}}}c^\dagger_{-\vec{\mathbf{k}}\downarrow}c^\dagger_{-,\uparrow}c^\dagger_{\vec{\mathbf{k}},\downarrow}(-u_{\vec{\mathbf{k}}}\phi_{\vec{\mathbf{k}}}+v_{\vec{\mathbf{k}}})+c_{-\vec{\mathbf{k}},\downarrow}(-u_{\vec{\mathbf{k}}}\phi_{\vec{\mathbf{k}}}+v_{\vec{\mathbf{k}}})
\end{aligned}$$

where we have used the fermionic anticommutation relations to swap $c_{\vec{\mathbf{k}},\uparrow}$ with $c^\dagger_{\vec{\mathbf{k}},\uparrow}$ before pulling it through the rest of the operators to annihilate the ground state. Thus, we see that these operators annihilate the ground state if we have

$$v_{\vec{\mathbf{k}}} = u_{\vec{\mathbf{k}}}\phi_{\vec{\mathbf{k}}}$$

In principle, we could use this and the requirement that the transformation stay canonical to determine the parameters $u_{\vec{\mathbf{k}}}$ and $v_{\vec{\mathbf{k}}}$. But we continue...

Next, we compute the commutator $[H, \alpha^\dagger_{\vec{\mathbf{k}}\uparrow}]$ directly. We have previously done a mean field expansion on the BCS Hamiltonian

$$H = \sum_{\vec{\mathbf{k}},\sigma}\epsilon_{\vec{\mathbf{k}}}c^\dagger_{\vec{\mathbf{k}},\sigma}c_{\vec{\mathbf{k}},\sigma} - \sum_{\vec{\mathbf{k}},\vec{\mathbf{k}}'}\frac{g}{V}c^\dagger_{\vec{\mathbf{k}}\uparrow}c^\dagger_{-\vec{\mathbf{k}}\downarrow}c_{\vec{\mathbf{k}}',\uparrow}c_{-\vec{\mathbf{k}}',\downarrow}$$

using the mean field $\Delta = -\frac{g}{V}\langle c_{\vec{\mathbf{k}},\uparrow} c_{-\vec{\mathbf{k}},\downarrow}\rangle$. After decomposing, we find that

$$\begin{aligned}
H &= \sum_{\vec{\mathbf{k}},\sigma} \epsilon_{\vec{\mathbf{k}}} c^\dagger_{\vec{\mathbf{k}},\sigma} c_{\vec{\mathbf{k}},\sigma} - \sum_{\vec{\mathbf{k}},\vec{\mathbf{k}}'} \frac{g}{V} c^\dagger_{\vec{\mathbf{k}}\uparrow} c^\dagger_{-\vec{\mathbf{k}}\downarrow} c_{\vec{\mathbf{k}}',\uparrow} c_{-\vec{\mathbf{k}}',\downarrow} \\
&= \sum_{\vec{\mathbf{k}},\sigma} \epsilon_{\vec{\mathbf{k}}} c^\dagger_{\vec{\mathbf{k}},\sigma} c_{\vec{\mathbf{k}},\sigma} - \sum_{\vec{\mathbf{k}},\vec{\mathbf{k}}'} \frac{V}{g} \left(\overline{\Delta} - \left(\overline{\Delta} - \frac{g}{V} c^\dagger_{\vec{\mathbf{k}}\uparrow} c^\dagger_{-\vec{\mathbf{k}}\downarrow}\right)\right)\left(\Delta - \left(\Delta - \frac{g}{V} c_{\vec{\mathbf{k}}',\uparrow} c_{-\vec{\mathbf{k}}',\downarrow}\right)\right) \\
&= \sum_{\vec{\mathbf{k}},\sigma} \epsilon_{\vec{\mathbf{k}}} c^\dagger_{\vec{\mathbf{k}},\sigma} c_{\vec{\mathbf{k}},\sigma} + \sum_{\vec{\mathbf{k}},\vec{\mathbf{k}}'} \left[\overline{\Delta} c_{-\vec{\mathbf{k}}\downarrow} c_{\vec{\mathbf{k}}\uparrow} + c^\dagger_{\vec{\mathbf{k}}\uparrow} c^\dagger_{-\vec{\mathbf{k}}\downarrow} \Delta\right] + \frac{V}{g}\overline{\Delta}\Delta
\end{aligned}$$

In principle, defining the Nambu spinor $\Psi_{\vec{\mathbf{k}}} = (c_{\vec{\mathbf{k}}\uparrow}, c^\dagger_{-\vec{\mathbf{k}}\downarrow})^T$, we could rewrite the Hamiltonian as (up to constants)

$$H = \begin{pmatrix} c^\dagger_{\vec{\mathbf{k}}\uparrow} \\ c_{-\vec{\mathbf{k}}\downarrow} \end{pmatrix} \cdot \begin{pmatrix} \epsilon_{\vec{\mathbf{k}}} & \Delta \\ \overline{\Delta} & -\epsilon_{\vec{\mathbf{k}}} \end{pmatrix} \begin{pmatrix} c_{\vec{\mathbf{k}}\uparrow} \\ c^\dagger_{-\vec{\mathbf{k}}\downarrow} \end{pmatrix} = \Psi^\dagger_{\vec{\mathbf{k}}} \begin{pmatrix} \epsilon_{\vec{\mathbf{k}}} & \Delta \\ \overline{\Delta} & -\epsilon_{\vec{\mathbf{k}}} \end{pmatrix} \Psi_{\vec{\mathbf{k}}}$$

which would give the desired result. However, we continue as requested: We compute the equation of motion

$$[H, \alpha^\dagger_{\vec{\mathbf{k}}\uparrow}] = [H, u_{\vec{\mathbf{k}}} c^\dagger_{\vec{\mathbf{k}}\uparrow} + v_{\vec{\mathbf{k}}} c_{-\vec{\mathbf{k}}\downarrow}] = E_{\vec{\mathbf{k}}} a^\dagger_{\vec{\mathbf{k}}\uparrow}$$

For the kinetic term, we find that

$$[H_{\text{kin}}, u_{\vec{\mathbf{k}}} c^\dagger_{\vec{\mathbf{k}}\uparrow} + v_{\vec{\mathbf{k}}} c_{-\vec{\mathbf{k}}\downarrow}] = \epsilon_{\vec{\mathbf{k}}} u_{\vec{\mathbf{k}}} c^\dagger_{\vec{\mathbf{k}}\uparrow} - \epsilon_{\vec{\mathbf{k}}} v_{\vec{\mathbf{k}}} c_{-\vec{\mathbf{k}}\downarrow}$$

and for the mean field term, we also find that

$$\begin{aligned}
[H_{\text{MF}}, v_{\vec{\mathbf{k}}} c_{-\vec{\mathbf{k}}\downarrow}] &= \Delta v_{\vec{\mathbf{k}}} [c^\dagger_{\vec{\mathbf{k}}\uparrow} c^\dagger_{-\vec{\mathbf{k}}\downarrow}, c_{-\vec{\mathbf{k}}\downarrow}] \\
&= \Delta v_{\vec{\mathbf{k}}} \left(c^\dagger_{\vec{\mathbf{k}}\uparrow} c^\dagger_{-\vec{\mathbf{k}}\downarrow} c_{-\vec{\mathbf{k}}\downarrow} - c_{-\vec{\mathbf{k}}\downarrow} c^\dagger_{\vec{\mathbf{k}}\uparrow} c^\dagger_{-\vec{\mathbf{k}}\downarrow}\right) \\
&= \Delta v_{\vec{\mathbf{k}}} \left(c^\dagger_{\vec{\mathbf{k}}\uparrow} c^\dagger_{-\vec{\mathbf{k}}\downarrow} c_{-\vec{\mathbf{k}}\downarrow} + c^\dagger_{\vec{\mathbf{k}}\uparrow}(-c^\dagger_{-\vec{\mathbf{k}}\downarrow} c_{-\vec{\mathbf{k}}\downarrow} + 1)\right) \\
&= \Delta v_{\vec{\mathbf{k}}} c^\dagger_{\vec{\mathbf{k}}\uparrow} \\
[H_{\text{MF}}, u_{\vec{\mathbf{k}}} c^\dagger_{\vec{\mathbf{k}}\uparrow}] &= \overline{\Delta} u_{\vec{\mathbf{k}}} [c_{-\vec{\mathbf{k}}\downarrow} c_{\vec{\mathbf{k}}\uparrow}, c^\dagger_{\vec{\mathbf{k}}\uparrow}] \\
&= \overline{\Delta} u_{\vec{\mathbf{k}}} \left(c_{-\vec{\mathbf{k}}\downarrow} c_{\vec{\mathbf{k}}\uparrow} c^\dagger_{\vec{\mathbf{k}}\uparrow} - c^\dagger_{\vec{\mathbf{k}}\uparrow} c_{-\vec{\mathbf{k}}\downarrow} c_{\vec{\mathbf{k}}\uparrow}\right) \\
&= \overline{\Delta} u_{\vec{\mathbf{k}}} \left(c_{-\vec{\mathbf{k}}\downarrow} c_{\vec{\mathbf{k}}\uparrow} c^\dagger_{\vec{\mathbf{k}}\uparrow} + c_{-\vec{\mathbf{k}}\downarrow}(-c_{\vec{\mathbf{k}}\uparrow} c^\dagger_{\vec{\mathbf{k}}\uparrow} + 1)\right) \\
&= \overline{\Delta} u_{\vec{\mathbf{k}}} c_{-\vec{\mathbf{k}}\downarrow}
\end{aligned}$$

Thus, we find that

$$[H, \alpha^\dagger_{\vec{\mathbf{k}}\uparrow}] = \epsilon_{\vec{\mathbf{k}}} u_{\vec{\mathbf{k}}} c^\dagger_{\vec{\mathbf{k}}\uparrow} - \epsilon_{\vec{\mathbf{k}}} v_{\vec{\mathbf{k}}} c_{-\vec{\mathbf{k}}\downarrow} + \Delta v_{\vec{\mathbf{k}}} c^\dagger_{\vec{\mathbf{k}}\uparrow} + \overline{\Delta} u_{\vec{\mathbf{k}}} c_{-\vec{\mathbf{k}}\downarrow}$$

By comparison of the above equation with the equation $[H, \alpha^\dagger_{\vec{\mathbf{k}}\uparrow}] = E_{\vec{\mathbf{k}}}\alpha^\dagger_{\vec{\mathbf{k}}\uparrow}$ and comparing the coefficients of $c^\dagger_{\vec{\mathbf{k}}\uparrow}$ and $c_{\vec{\mathbf{k}}\uparrow}$, we find that $(u_{\vec{\mathbf{k}}}, v_{\vec{\mathbf{k}}})$ is an eigenvector of the desired matrix. □

Problem 19. One useful way to regard superconductors is via the semiconductor analogy, in which the quasiparticles are treated like the positive and negative energy excitations of a semiconductor. To do so we divide the Brillouin zone up into equal halves and redefine a set of positive and negative energy quasiparticle operators according to $\alpha^\dagger_{\mathbf{k},\sigma+} = a^\dagger_{\mathbf{k}\sigma}$ and $\alpha^\dagger_{\mathbf{k}\sigma-} = \mathrm{sgn}(\sigma)\, a_{-\mathbf{k}-\sigma}$ for $\mathbf{k} \in \frac{1}{2}\mathrm{BZ}$.

- By this procedure rewrite the BCS Hamiltonian in terms of these new operators, and show that the excitation spectrum can be interpreted in terms of an empty band of positive energy excitation and a filled band of negative energy excitations.
- Show that the BCS ground state wave functions can be regarded as a filled sea of negative energy quasiparticle states and an empty sea of positive energy quasiparticle states.

Proof. For reference the BCS Hamiltonian referred to in the problem is the already diagonalised BCS mean field Hamiltonian in the quasiparticle basis

$$H = \sum_{\vec{\mathbf{k}}} E_{\vec{\mathbf{k}}}(a^\dagger_{\vec{\mathbf{k}}\uparrow}a_{\vec{\mathbf{k}}\uparrow} - a_{-\vec{\mathbf{k}}\downarrow}a^\dagger_{-\vec{\mathbf{k}}\downarrow}) + V\frac{\overline{\Delta}\Delta}{g_0}.$$

We ignore the constants and divide the sum to a sum over half of the Brillouin zone to find

$$\begin{aligned} H &= \sum_{\vec{\mathbf{k}}\in\frac{1}{2}\mathrm{BZ}} E_{\vec{\mathbf{k}}}(a^\dagger_{\vec{\mathbf{k}}\uparrow}a_{\vec{\mathbf{k}}\uparrow} - a_{-\vec{\mathbf{k}}\downarrow}a^\dagger_{-\vec{\mathbf{k}}\downarrow}) + \sum_{\vec{\mathbf{k}}\in\frac{1}{2}\mathrm{BZ}} E_{\vec{\mathbf{k}}}(a^\dagger_{-\vec{\mathbf{k}}\uparrow}a_{-\vec{\mathbf{k}}\uparrow} - a_{\vec{\mathbf{k}}\downarrow}a^\dagger_{\vec{\mathbf{k}}\downarrow}) \\ &= \sum_{\vec{\mathbf{k}}\in\frac{1}{2}\mathrm{BZ},\sigma} E_{\vec{\mathbf{k}}}(a^\dagger_{\vec{\mathbf{k}}\sigma}a_{\vec{\mathbf{k}}\sigma} - a_{-\vec{\mathbf{k}}\sigma}a^\dagger_{-\vec{\mathbf{k}}\sigma}) \\ &= \sum_{\vec{\mathbf{k}}\in\frac{1}{2}\mathrm{BZ},\sigma} E_{\vec{\mathbf{k}}}\left(\alpha^\dagger_{\vec{\mathbf{k}}\sigma+}\alpha_{\vec{\mathbf{k}}\sigma+} - \alpha^\dagger_{\vec{\mathbf{k}}-}\alpha_{\vec{\mathbf{k}}\sigma-}\right) \end{aligned}$$

from which we see that the Hamiltonian has factored into positive and negative energy quasiparticles.

We previously the BCS ground state wavefunction in terms of the Bogoliubov quasiparticles by annihilating all quasiparticles

$$|BCS\rangle = \sum_{\vec{\mathbf{k}}} a_{-\vec{\mathbf{k}}\downarrow} a_{\vec{\mathbf{k}}\uparrow} |0\rangle .$$

Dividing the Brillouin zone again into two, we find that

$$\begin{aligned} |BCS\rangle &= \prod_{\vec{\mathbf{k}}\in\frac{1}{2}\mathrm{BZ}} (a_{\vec{\mathbf{k}}\downarrow} a_{\vec{\mathbf{k}}\uparrow})(a_{-\vec{\mathbf{k}}\downarrow} a_{-\vec{\mathbf{k}}\uparrow}) |0\rangle \\ &= \prod_{\vec{\mathbf{k}}\in\frac{1}{2}\mathrm{BZ},\sigma} \alpha_{\vec{\mathbf{k}}\sigma+} \prod_{\vec{\mathbf{k}}\in\frac{1}{2}\mathrm{BZ},\sigma} \alpha^{\dagger}_{\vec{\mathbf{k}}\sigma-} |0\rangle \end{aligned}$$

from which we see that the negative energy quasiparticle band is fully filled, while the positive energy quasiparticle band is fully empty. □

Problem 20.

- Assuming that the Debye frequency is a small fraction of the bandwidth, show that the difference between the superconducting and normal-state free energies can be written as

$$\mathcal{F}_S - \mathcal{F}_N = -2TN(0) \int_{-\omega_D}^{\omega_D} \ln \left[\frac{\cosh\left(\sqrt{\varepsilon^2 + |\Delta|^2}/2T\right)}{\cosh(\varepsilon/2T)} \right] d\varepsilon + V \frac{|\Delta|^2}{g_0}. \tag{9}$$

  Notice that the previous expression is invariant under changes in the phase of the gap parameter. What consequences does it imply?

- By differentiating the expression of the Free energy with respect to Δ, confirm the zero temperature gap equation

$$\frac{V}{g_0 N(0)} = \int_0^{\omega_D} \frac{d\varepsilon}{\sqrt{\varepsilon^2 + \Delta_0^2}}, \tag{10}$$

  where $\Delta_0 = \Delta(T = 0)$. Use this result to eliminate g_0, to show that the free energy can be written as

$$\mathcal{F}_S - \mathcal{F}_N = N(0)\Delta_0^2 \Phi\left[\frac{\Delta}{\Delta_0}, \frac{T}{\Delta_0}\right], \tag{11}$$

  where the dimensionless function

$$\Phi[\delta, t] = \int_0^{\infty} dx \left\{ -4t \ln \frac{\cosh\left(\sqrt{x^2 + \delta^2}/2t\right)}{\cosh(x/2t)} + \frac{\delta^2}{\sqrt{x^2 + 1}} \right\}. \tag{12}$$

  What implies the change in the integration boundary to infinity?

Proof. The expression for the free energy is assumed to be known in the question: It can be derived by performing a Hubbard-Stratonovich transformation, integrating out the fermions and then performing a mean field substitution on the Hubbard-Stratonovich field, with which the path integral disappears and only the product of eigenvalues forming a determinant remain.

The free energy is given by

$$\mathcal{F}[\Delta, T] = -T \sum_{\vec{\mathbf{k}},n} \ln[\omega_n^2 + \epsilon_{\vec{\mathbf{k}}}^2 + |\Delta|^2] + V \frac{|\Delta|^2}{g_0}$$

where it is to be understood that the momentum sum is restricted to a shell around the Fermi surface - while the free energy would involve all electrons, not just those in this small shell, it makes no difference when finding minima, as they are noninteracting and hence behave like free electrons. In the path integral, this would be understood as simply integrating out these fermions and ignoring the constants resulting therefrom. Carrying out the Matsubara sum, it can be found that (as I understand, this result is also assumed)

$$\mathcal{F} = -2TV \int_{|\epsilon_{\vec{\mathbf{k}}}|<\omega_D} \frac{\mathrm{d}^3 k}{(2\pi)^3} \left[\ln[2\cosh(\beta E_{\vec{\mathbf{k}}}/2)]\right] + V \frac{|\Delta|^2}{g_0}$$

where $E_{\vec{\mathbf{k}}}$ is the quasiparticle dispersion $E_{\vec{\mathbf{k}}} = \sqrt{\epsilon_{\vec{\mathbf{k}}}^2 + \Delta^2}$ (assuming the gap parameter Δ is real).

The normal state free energy is just the free energy evaluated at zero gap parameter Δ. Substituting, we easily find that

$$\mathcal{F}_S - \mathcal{F}_N = -2TV \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \ln \left[\frac{\cosh\left(\sqrt{\varepsilon^2 + |\Delta|^2}/2T\right)}{\cosh(\varepsilon/2T)} \right] + V \frac{|\Delta|^2}{g_0}$$

The last step in this derivation is to replace the integration measure with an integral over the energy and the density of states $V \frac{\mathrm{d}^3 k}{(2\pi)^3} = N(\epsilon)\, \mathrm{d}\epsilon$ and approximate the density of states as constant within a tiny shell around the Fermi surface to get

$$\mathcal{F}_S - \mathcal{F}_N = -2TN(0) \int_{-\omega_D}^{\omega_D} \ln \left[\frac{\cosh\left(\sqrt{\varepsilon^2 + |\Delta|^2}/2T\right)}{\cosh(\varepsilon/2T)} \right] d\varepsilon + V \frac{|\Delta|^2}{g_0}.$$

where $N(0)$ is the density of states at the Fermi surface.

Next, we differentiate the free energy as requested to find

$$\frac{\mathrm{d}}{\mathrm{d}\Delta} \Delta\mathcal{F} = -2TN(0) \int_{-\omega_D}^{\omega_D} \tanh\left(\sqrt{\varepsilon^2 + |\Delta|^2}/2T\right) \frac{\Delta}{2T\sqrt{\epsilon^2 + \Delta^2}}$$

$$+\frac{2V\Delta}{g_0}=0$$

Noting that $\tanh(1/T)\to 1$ as $T\to 0$, we have

$$N(0)\int_{-\omega_D}^{\omega_D}\frac{1}{\sqrt{\epsilon^2+\Delta_0^2}}\,\mathrm{d}\epsilon=\frac{2V}{g_0}$$

Using the evenness of the integrand to allow us to integrate over only half of the specified interval, this reduces to

$$\frac{V}{g_0N(0)}=\int_0^{\omega_D}\frac{\mathrm{d}\varepsilon}{\sqrt{\varepsilon^2+\Delta_0^2}}$$

as desired.

The final expression we are asked to show follows by simply substituting to find

$$\begin{aligned}\mathcal{F}_S-\mathcal{F}_N&=-2TN(0)\int_{-\omega_D}^{\omega_D}\ln\left[\frac{\cosh\left(\sqrt{\varepsilon^2+|\Delta|^2}/2T\right)}{\cosh(\varepsilon/2T)}\right]\mathrm{d}\varepsilon+|\Delta|^2N(0)\int_0^{\omega_D}\frac{\mathrm{d}\varepsilon}{\sqrt{\varepsilon^2+\Delta_0^2}}\\&=N(0)\int_0^{\omega_D}-4T\ln\left[\frac{\cosh\left(\sqrt{\varepsilon^2+|\Delta|^2}/2T\right)}{\cosh(\varepsilon/2T)}\right]+\frac{|\Delta|^2}{\sqrt{\epsilon^2+\Delta_0^2}}\,\mathrm{d}\epsilon\end{aligned}$$

The derivation of the desired expression

$$\mathcal{F}_S-\mathcal{F}_N=N(0)\Delta_0^2\,\Phi\left[\frac{\Delta}{\Delta_0},\frac{T}{\Delta_0}\right]$$

is now obvious. We note that during this derivation, we would replace ϵ/Δ_0 with x; in the case where the gap parameter is small, this would extend the integration bounds to infinity. □

Problem 21. Consider two superconductors which are connected via a small bridge, also superconducting, of length L. Each of those superconductors can be thought as a lead with all of its properties well defined and each of the wave function in this bridge

$$\Psi(x\to 0)=\Psi_1=|\Psi_0|e^{i\phi_1} \tag{13}$$

$$\Psi(x\to L)=\Psi_2=|\Psi_0|e^{i\phi_2} \tag{14}$$

for the edges. In this area the wave function can be thought as a superposition of those two

$$\Psi(x)=|\Psi_0|e^{i\phi_1}g(x)+|\Psi_0|e^{i\phi_2}(1-g(x)), \tag{15}$$

with the only requirement that $g(x \to 0) \to 1$ and $g(x \to L) \to 0$. We aim to get the behaviour of $g(x)$, in order to do so use the Ginzburg Landau equation

$$-\frac{\hbar^2}{2m}\frac{\partial^2 \Psi}{\partial x^2} + \alpha\Psi + \beta|\Psi|^2\Psi = 0, \tag{16}$$

to have a complete description.

Hint: It might be useful to think of $\Psi(x) = |\Psi_0| f(x)$ and rewrite an equation for $f(x)$. Use $|\Psi_0|^2 = -\alpha/\beta$ and define also $\xi^2 = \hbar^2/(2m|\alpha|)$. Use the hypothesis of short link $L \ll \xi$.

Proof. As suggested, we think of $\Psi(x)$ as $|\Psi_0| f(x)$. Then, we substitute this into the Ginzburg-Landau equation to find

$$\begin{aligned} 0 &= -\frac{\hbar^2}{2m}\frac{\partial^2 f(x)}{\partial x^2} + \alpha f(x) + \beta|\Psi_0|^2|f(x)|^2 f(x) \\ &= -\frac{\hbar^2}{2m}\frac{\partial^2 f(x)}{\partial x^2} + \alpha f(x) - \alpha|f(x)|^2 f(x) \\ &= \zeta^2\frac{\partial^2 f(x)}{\partial x^2} + f(x)(1 - |f(x)|^2) \end{aligned}$$

Since we know L is small, the second derivative scales as

$$|f''| \sim \frac{\Delta f}{L^2}$$

Since $L \ll \zeta$, we find that

$$\zeta^2 f'' \sim \left(\frac{\zeta}{L}\right)^2 \Delta f \gg \Delta f$$

Hence, we find that

$$\zeta^2 f'' \approx 0$$

This was also to be expected, because we could have sent ζ^2 to infinity while keeping the magnitude of f constant (due to boundary conditions). Thus, we find that f is linear, $f = A + Bx$. There is only one solution that satisfies the boundary conditions:

$$f(x) = e^{i\phi_1} + \frac{x}{L}(e^{i\phi_2} - e^{i\phi_1}). \qquad \square$$

Problem 22. With the previous result, compute the Josephson current. Show that it depends exclusively on $\phi = \phi_2 - \phi_1$ and not on each phase individually. What is the expression for the free energy?

Proof. The current is given by

$$
\begin{aligned}
J_s &= \frac{e\hbar}{2mi}\left(\psi^*\frac{\partial\psi}{\partial x} - \psi\frac{\partial\psi^*}{\partial x}\right) \\
&= \frac{e\hbar}{m}\,\mathrm{Im}\left(\psi^*\frac{\partial\psi}{\partial x}\right) \\
&= \frac{e\hbar}{m}\,\mathrm{Im}\left[\left(e^{-i\phi_1} + \frac{x}{L}(e^{-i\phi_2} - e^{-i\phi_1})\right)\left(e^{i\phi_2} - e^{i\phi_1}\right)\right] \\
&= \frac{e\hbar}{m}\,\mathrm{Im}\left[e^{-i\phi_1}\left(e^{i\phi_2} - e^{i\phi_1}\right) + \frac{x}{L}\underbrace{\left(e^{-i\phi_2} - e^{-i\phi_1}\right)\left(e^{i\phi_2} - e^{i\phi_1}\right)}_{\mathbb{R}}\right] \\
&= \frac{e\hbar}{m}\,\mathrm{Im}\left[e^{-i\phi_1}\left(e^{i\phi_2} - e^{i\phi_1}\right)\right] \\
&= \frac{e\hbar}{m}\,\mathrm{Im}\left[e^{i(\phi_2-\phi_1)} - 1\right]
\end{aligned}
$$

at which point we have already shown that it only depends on the difference. However, we can simplify it further to find

$$
J_s = \frac{e\hbar}{m}\sin(\phi_2 - \phi_1)
$$

The Ginzburg-Landau free energy is given by

$$
F_{GL} = \int \frac{\hbar^2}{2m}|\boldsymbol{\nabla}\Psi|^2 + \alpha|\Psi|^2 + \beta|\Psi|^4\,\mathrm{d}V
$$

In the short junction limit, we approximate the magnitude of Ψ as being constant and equal to $|\Psi_0|$. Hence, the other terms vanish and we are left with

$$
\begin{aligned}
F_{GL} &= A\int \frac{\hbar^2}{2m}|\boldsymbol{\nabla}\Psi|^2\,\mathrm{d}x \\
&= \frac{\hbar^2 A}{2m}\int\left|\frac{\partial\Psi}{\partial x}\right|^2\mathrm{d}x \\
&= \frac{\hbar^2 A|\Psi_0|^2}{2mL^2}\int_0^L\left|e^{i\phi_2} - e^{i\phi_1}\right|^2\mathrm{d}x \\
&= \frac{\hbar^2 A|\Psi_0|^2}{2m}\left[(\cos\phi_2 - \cos\phi_1)^2 + (\sin\phi_2 - \sin\phi_1)^2\right] \\
&= \frac{\hbar^2 A|\Psi_0|^2}{2m}\left[2 - 2\cos\phi_1\cos\phi_2 - 2\sin\phi_1\sin\phi_2\right] \\
&= \frac{\hbar^2 A|\Psi_0|^2}{2m}\left[1 - \cos(\phi_2 - \phi_1)\right]
\end{aligned}
$$

where A is the area of the superconductor. □

Problem 23. How is the current changed when one applies a voltage? Consider the Schrödinger equation and assume $\Psi(x,t) = |\Psi(x)|e^{i\phi(t)}$. Use that to find $\phi(t)$ assuming $\phi(t=0) = \phi_0$. You may associate the energy of the external superconductors with their chemical potential, and the difference of chemical potential with the applied voltage.

Proof. We assume that the applied voltage is small such that the magnitude does not change, i.e. $\Psi(x,t) = |\Psi(x)|e^{i\phi(t)}$. Now, we use the Schrödinger equation

$$-i\hbar\frac{\partial\psi}{\partial t} = -i\hbar\left[i\phi'(t)|\Psi(x)|e^{i\phi(t)}\right] = \phi'(t)|\Psi(x)|e^{i\phi(t)} = H\psi$$

Since the Hamiltonian couples to the order parameter directly as a chemical potential, we find that

$$\hbar\phi'(t)|\Psi(x)|e^{i\phi(t)} = \mu|\Psi(x)|e^{i\phi(t)}$$

which obviously has the solution

$$\phi(t) = \phi_0 + \frac{\mu}{\hbar}t$$

We recall from earlier that

$$J_s = \frac{e\hbar}{m}\sin(\phi_2 - \phi_1).$$

We can set $\phi_1(t) = \phi_1(0) = 0$ without loss of generality. Thus, we will assume that there is only a voltage acting on ϕ_2. This leads to the solution (setting the chemical potential $\mu = eV$)

$$J_s = \frac{e\hbar}{m}\sin\left(\phi_0 + \frac{eV}{\hbar}t\right)$$

Note that if there is a voltage $-V$ acting on the phase ϕ_1, then there will be an additional factor of 2 as the voltage difference is double. □